

SAMSUNG BIOLOGICS Co., Ltd.

Financial Statements

December 31, 2018 and 2017

(With Independent Auditors' Report Thereon)

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Independent Auditors' Report

Based on a report originally issued in Korean

To the Shareholders and Board of Directors of
Samsung Biologics Co., Ltd.

Opinion

We have audited the financial statements of Samsung Biologics Co., Ltd. ("the Company"), which comprise the statements of financial position as of December 31, 2018 and 2017, the statements of comprehensive income, changes in equity and cash flows for the years then ended, and notes, comprising significant accounting policies and other explanatory information.

In our opinion, the accompanying financial statements present fairly, in all material respects, the financial position of the Company as of December 31, 2018 and 2017, and its financial performance and its cash flows for the years then ended in accordance with Korean International Financial Reporting Standards ("K-IFRS").

Basis for Opinion

We conducted our audits in accordance with Korean Standards on Auditing (KSAs). Our responsibilities under those standards are further described in the Auditors' Responsibilities for the Audit of the Financial Statements section of our report. We are independent of the Company in accordance with the ethical requirements that are relevant to our audit of the financial statements in Republic of Korea, and we have fulfilled our other ethical responsibilities in accordance with these requirements. We believe that the audit evidence we have obtained is sufficient and appropriate to provide a basis for our opinion

Emphasis of Matter

Without qualifying our opinion, we draw attention to following:

1. Exercised a call option

As noted in Note 14, Note 30 and Note 34 to the financial statements, Biogen Therapeutics Inc. exercised a call option on June 29, 2018 to buy a 50%-1 share in Samsung Bioepis, the Company's investee. As a result, the Company's ownership in Samsung Bioepis is 50%+1 shares as of December 31, 2018.

2. Securities & Futures Commission's ruling and administrative litigation

As described in Note 35 to the financial statements, on July 12, 2018, the Securities and Futures Commission (the "SFC") resolved to take an action in connection with the fact that the Company omitted to disclose the relevant notes to the financial statements regarding the joint venture agreement in accordance with the Act on External Audit. Further, on November 14, 2018, the SFC resolved to take an action in connection with the error on accounting treatment related to consolidation scope in accordance with the Act on External Audit.

On November 27, 2018 and December 18, 2018, the Company submitted an application to the Seoul Administrative Court, respectively, for the suspension of the action for restatement of the financial statements, and obtained the notification from Seoul Administrative Court that the effective of decision by SFC would be suspended until 30 days after the final ruling of litigation between the Company and SFC on January 22, 2019 and February 19, 2019, respectively. Based on this, the Company did not restate its financial statements.

The outcome of the outstanding administrative litigation is difficult to predict, but if the Company loses the administrative litigation, the financial statements would be restated as described in Note 35 to the financial statements.

Key Audit Matters

Key audit matters are those matters that, in our professional judgment, were of most significance in our audit of the financial statements as of and for the year ended December 31, 2018. These matters were addressed in the context of our audit of the financial statements as a whole, and in forming our opinion thereon, and we do not provide a separate opinion on these matters.

1. Assessment of investments in associates and joint ventures

The carrying amount of the investments in associates and joint ventures(the "investments in associates") as of December 31, 2018 is ₩ 2,561,803 million. As noted in Note 4 to the financial statements, the Company reviews whether there are any impairment indicators of investments in associates and joint ventures annually at the end of each reporting period and performs an impairment test for investments whose impairment indicators are identified. The recoverable amount of investments in associates and joint ventures is the greater of its value in use and its fair value less costs to sell.

As inherent uncertainties and management bias are involved in key assumptions such as sales growth rate, discount rate and terminal growth rate that have a significant impact on estimating the value of investments, we have identified the impairment test for investment in related businesses as a key audit matter.

The primary procedures we performed to address this key audit matter included the following.

- ① Assessing the independence and qualification of the external valuation specialists engaged by the Company to evaluate the value-in-use
- ② Engaging of our internal valuation specialists, to assess the appropriateness of key assumptions, such as sales growth rate, discount rate and terminal growth rate applied to calculate the value-in-use
- ③ Assessing the reasonableness of the key assumptions and variables used in assessment of the value-in-use by comparing the latest business plans adopted by management with historical performance and available external data
- ④ Comparing the actual results for the current year with management's estimates in their cash flows calculations for the previous year in order to assess the historical accuracy of the management's forecasting process

2. Revenue recognition

The Company focuses on operation of a CMO(Contract Manufacturing Organization) business entrusted with the production of biopharmaceuticals, and the Company's revenue recognition process includes technology transfer procedures, regulatory approval procedures, commercial production procedures. The management of the Company determines the amount and timing of revenue recognition in need of significant judgement to meet the requirements of K-IFRS No. 1115. In consideration of significance of related incomes in financial statements and complexity of management judgement, the Company identified the revenue recognition as a key audit matters.

The primary procedures we performed to address this key audit matter included the following

- ① Evaluating certain Internal controls over the determination of the amount and timing of revenue recognition
- ② Assessing the appropriateness of the effects of changes in basic retained earnings from the introduction of K-IFRS No.1115
- ③ Inspecting contracts for each project to assess the appropriateness of revenue recognition accounting
- ④ Performing independent recalculation of progress by project and contract assets and contract liabilities

Other Matter

The procedures and practices utilized in the Republic of Korea to audit such financial statements may differ from those generally accepted and applied in other countries.

Responsibilities of Management and Those Charged with Governance for the Financial Statements

Management is responsible for the preparation and fair presentation of the financial statements in accordance with K-IFRS, and for such internal control as management determines is necessary to enable the preparation of financial statements that are free from material misstatement, whether due to fraud or error.

In preparing the financial statements, management is responsible for assessing the company's ability to continue as a going concern, disclosing, as applicable, matters related to going concern and using the going concern basis of accounting unless management either intends to liquidate the company or to cease operations, or has no realistic alternative but to do so.

Those charged with governance are responsible for overseeing the company's financial reporting process.

Auditors' Responsibilities for the Audit of the Financial Statements

Our objectives are to obtain reasonable assurance about whether the financial statements as a whole are free from material misstatement, whether due to fraud or error, and to issue an auditors' report that includes our opinion. Reasonable assurance is a high level of assurance, but is not a guarantee that an audit conducted in accordance with KSAs will always detect a material misstatement when it exists. Misstatements can arise from fraud or error and are considered material if, individually or in the aggregate, they could reasonably be expected to influence the economic decisions of users taken on the basis of these financial statements.

As part of an audit in accordance with KSAs, we exercise professional judgment and maintain professional skepticism throughout the audit. We also:

- ✓ Identify and assess the risks of material misstatement of the financial statements, whether due to fraud or error, design and perform audit procedures responsive to those risks, and obtain audit evidence that is sufficient and appropriate to provide a basis for our opinion. The risk of not detecting a material misstatement resulting from fraud is higher than for one resulting from error, as fraud may involve collusion, forgery, intentional omissions, misrepresentations, or the override of internal control.
- ✓ Obtain an understanding of internal control relevant to the audit in order to design audit procedures that are appropriate in the circumstances, but not for the purpose of expressing an opinion on the effectiveness of the Company's internal control.
- ✓ Evaluate the appropriateness of accounting policies used in the preparation of the financial statements and the reasonableness of accounting estimates and related disclosures made by management.
- ✓ Conclude on the appropriateness of management's use of the going concern basis of accounting and, based on the audit evidence obtained, whether a material uncertainty exists related to events or conditions that may cast significant doubt on the Company's ability to continue as a going concern. If we conclude that a material uncertainty exists, we are required to draw attention in our auditors' report to the related disclosures in the financial statements or, if such disclosures are inadequate, to modify our opinion. Our conclusions are based on the audit evidence obtained up to the date of our auditors' report. However, future events or conditions may cause the Company to cease to continue as a going concern.
- ✓ Evaluate the overall presentation, structure and content of the financial statements, including the disclosures, and whether the financial statements represent the underlying transactions and events in a manner that achieves fair presentation.

We communicate with those charged with governance regarding, among other matters, the planned scope and timing of the audit and significant audit findings, including any significant deficiencies in internal control that we identify during our audit.

We also provide those charged with governance with a statement that we have complied with relevant ethical requirements regarding independence, and communicate with them all relationships and other matters that may reasonably be thought to bear on our independence, and where applicable, related safeguards.

From the matters communicated with those charged with governance, we determine those matters that were of most significance in the audit of the financial statements of the current period and are therefore the key audit matters. We describe these matters in our auditors' report unless law or regulation precludes public disclosure about the matter or when, in extremely rare circumstances, we determine that a matter should not be communicated in our report because the adverse consequences of doing so would reasonably be expected to outweigh the public interest benefits of such communication.

KPMG Samjong Accounting Corp.
Seoul, Korea
March 14, 2019

This report is effective as of March 14, 2019, the audit report date. Certain subsequent events or circumstances, which may occur between the audit report date and the time of reading this report, could have a material impact on the accompanying financial statements and notes thereto. Accordingly, the readers of the audit report should understand that the above audit report has not been updated to reflect the impact of such subsequent events or circumstances, if any.
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Samsung Biologics Co., Ltd.
Statements of Financial Position
As of December 31, 2018 and 2017

(In thousands of won)

	Note	2018	2017
Assets:			
Cash and cash equivalents	6,7,33 ₩	17,621,905	358,088,091
Short-term financial instruments	6,33	550,000,000	-
Trade and other receivables	6,9,31,33	121,733,350	25,817,150
Inventories	3,11,25	302,360,715	209,986,717
Deposits provided	6,12,33	560,960,126	22,605,000
Other current assets	13	5,508,443	4,498,177
Contract assets	3,10,25	19,499,176	-
Current assets		<u>1,577,683,715</u>	<u>620,995,135</u>
Long-term financial instruments	6,8,33	7,000	7,000
Investments in associates and joint ventures	14,30,31	2,561,802,521	5,008,234,005
Property, plant and equipment	15,31	1,717,720,988	1,513,332,618
Intangible assets	16	19,981,634	18,999,003
Deposits provided	6,12,33	24,230,355	20,560,000
Other non-current assets	13	1,902,973	963,402
Contract assets	10	77,079,462	-
Non-current assets		<u>4,402,724,933</u>	<u>6,562,096,028</u>
Total assets	₩	<u>5,980,408,648</u>	<u>7,183,091,163</u>
Liabilities:			
Trade and other payables	6,18,21,31,33 ₩	170,338,294	158,214,846
Derivative liabilities	6	-	1,933,562,427
Emission liability	19	1,493,982	502,080
Bonds and borrowings	6,17,30,33	386,583,355	138,676,578
Other current liabilities	20	87,695,614	56,486,175
Contract liabilities	10	24,794,825	-
Current tax liabilities		39,623,717	-
Current liabilities		<u>710,529,787</u>	<u>2,287,442,106</u>
Trade and other payables	6,18,21,31,33	3,988,729	2,650,638
Bonds and borrowings	6,17,30,33	486,112,752	407,235,570
Derivative liabilities	6	190,330	-
Defined benefit liabilities	21	13,266,305	3,996,064
Contract liabilities	3,10,31	76,309,965	-
Deferred tax liabilities	28	534,542,329	505,315,892
Non-current liabilities		<u>1,114,410,410</u>	<u>919,198,164</u>
Total liabilities		<u>1,824,940,197</u>	<u>3,206,640,270</u>
Equity:			
Share capital	1,22	165,412,500	165,412,500
Share premium	22	2,487,313,082	2,487,313,082
Accumulated other comprehensive income	14,22	(6,975,961)	(4,009,446)
Retained earnings	22	1,509,718,830	1,327,734,757
Total equity		<u>4,155,468,451</u>	<u>3,976,450,893</u>
Total liabilities and equity	₩	<u>5,980,408,648</u>	<u>7,183,091,163</u>

See accompanying notes to the financial statements.

Samsung Biologics Co., Ltd.
Statements of Comprehensive Income
For the years ended December 31, 2018 and 2017

(In thousands of won)

	Note	2018	2017
Revenue	3,5,23,31 ₩	535,805,801	464,629,334
Cost of revenue	3,11,19,23, 25,31	(390,645,859)	(332,829,273)
Gross profit		<u>145,159,942</u>	<u>131,800,061</u>
Selling, general and administrative expenses	24,25,31	(89,481,197)	(65,820,659)
Operating profit		<u>55,678,745</u>	<u>65,979,402</u>
Other income	26,31	390,399,125	889,983
Other expenses	26	(8,748,273)	(136,243)
Finance income	6,27	19,320,789	23,219,775
Finance costs	6,27	(24,337,217)	(76,345,332)
Equity loss on investments in associates and joint ventures	14	(129,337,333)	(129,602,338)
Profit (loss) before income tax		<u>302,975,836</u>	<u>(115,994,753)</u>
Income tax expense (benefit)	28	78,866,642	(19,022,637)
Net profit (loss) for the period		<u>224,109,194</u>	<u>(96,972,116)</u>
Other comprehensive loss		(2,966,514)	(8,956,451)
Items that will never be reclassified to profit or loss:			
Defined benefit plan remeasurements	21	(3,930,107)	(2,189,613)
Equity adjustments arising from investments in associates and joint ventures	14,28	(663,926)	682,446
Items that are or may be reclassified to profit or loss:			
Equity adjustments arising from investments in associates and joint ventures	14,28	1,627,519	(7,449,284)
Total comprehensive income (loss) for the period	₩	<u>221,142,680</u>	<u>(105,928,567)</u>
Earnings (loss) per share			
Basic and diluted earnings (loss) per share (in won)	29 ₩	3,387	(1,466)

See accompanying notes to the financial statements

Samsung Biologics Co., Ltd.
Statements of Changes in Equity
For the years ended December 31, 2018 and 2017

(In thousands of won)

	Share capital	Share premium	Accumulated other comprehensive income (loss)	Retained earnings	Total equity
Balance as of January 1, 2017	₩ 165,412,500	2,487,313,082	4,947,005	1,424,706,873	4,082,379,460
Comprehensive loss for the year					
Net loss for the year	-	-	-	(96,972,116)	(96,972,116)
Defined benefit plan remeasurements	-	-	(2,189,613)	-	(2,189,613)
Equity adjustments arising from investments in associates and joint ventures	-	-	(6,766,838)	-	(6,766,838)
Total comprehensive loss for the year	-	-	(8,956,451)	(96,972,116)	(105,928,567)
Balance as of December 31, 2017	165,412,500	2,487,313,082	(4,009,446)	1,327,734,757	3,976,450,893
Balance as of January 1, 2018	165,412,500	2,487,313,082	(4,009,446)	1,327,734,757	3,976,450,893
Adjustment on initial application of K-IFRS No.1115 (net of tax) (Note 3)	-	-	-	(41,974,909)	(41,974,909)
Adjusted balance as of January 1, 2018	165,412,500	2,487,313,082	(4,009,446)	1,285,759,848	3,934,475,984
Comprehensive income for the year					
Net profit for the year	-	-	-	224,109,195	224,109,195
Defined benefit plan remeasurements	-	-	(3,930,107)	-	(3,930,107)
Equity adjustments arising from investments in associates and joint ventures	-	-	640,127	-	640,127
Disposal of investments in associates and joint ventures	-	-	323,465	(150,213)	173,252
Total comprehensive income for the year	-	-	(2,966,515)	223,958,982	220,992,467
Balance as of December 31, 2018	₩ 165,412,500	2,487,313,082	(6,975,961)	1,509,718,830	4,155,468,451

See accompanying notes to the financial statements.

Samsung Biologics Co., Ltd.
Statements of Cash Flows
For the years ended December 31, 2018 and 2017

(In thousands of won)

	2018	2017
Cash flows from operating activities		
Net profit (loss) for the period	₩ 224,109,195	(96,972,116)
Adjustments for:		
Retirement benefits	10,070,884	7,637,599
Depreciation	85,522,362	64,188,480
Amortization of intangible assets	5,928,193	5,737,747
Reversal of loss on valuation of inventories	(1,932,844)	(2,038,561)
Impairment loss on contract assets	18,641,061	-
Gain on disposal of property, plant and equipment	(12,821)	-
Loss on disposal of property, plant and equipment	614,509	4,946
Impairment loss on intangible assets	10,756	-
Loss on disposal of intangible assets	-	39,600
Gain on foreign currency translations	(112,646)	(3,512,447)
Loss on foreign currency translations	3,291,913	331,259
Gain on foreign currency transactions	-	(6,830,890)
Loss on valuation of derivatives	190,330	59,091,484
Gain on disposal of investments in associates and joint ventures	(389,213,672)	-
Cost of emissions	1,493,982	502,080
Gain from assets contributed	(45,800)	-
Interest income	(9,651,185)	(9,949,872)
Gain on valuation of financial assets at fair value through profit	(80,517)	-
Gain on disposal of financial assets at fair value through profit	(301,927)	-
Interest expenses	15,838,589	5,683,983
Equity loss on investments in associates and joint ventures	129,337,333	129,602,338
Income tax expense (benefit)	78,866,642	(19,022,637)
	<u>(51,544,858)</u>	<u>231,465,109</u>
Changes in:		
Trade receivables	(94,528,033)	47,807,078
Other receivables	(1,263,764)	(536,295)
Other current assets	(368,023)	1,065,034
Inventories	(90,441,155)	(43,728,176)
Deposits provided	(495,032)	(5,400,700)
Other non-current assets	(1,598,129)	(990,145)
Contract assets	(73,070,459)	-
Trade payables	(13,315,814)	13,180,780
Other payables	85,585,333	26,582,183
Other current liabilities	31,209,439	(3,837,616)
Other non-current payables	2,863,551	1,059,873
Contract liabilities	43,664,407	-
Emission liability	(502,080)	-
Retirement benefits paid	(906,053)	(428,339)
Transfer from affiliates of defined benefit liabilities	30,574	89,100
Contributions to plan assets	(6,638,266)	(13,085,879)
	<u>(119,773,504)</u>	<u>21,776,898</u>
Cash generated from operating activities	<u>52,790,833</u>	<u>156,269,891</u>
Interest received	8,486,910	10,433,121
Interest paid	(23,359,097)	(17,036,196)
Income tax refunded (paid)	(1,131,338)	149,511
Net cash from operating activities	<u>₩ 36,787,308</u>	<u>149,816,327</u>

See accompanying notes to the financial statements.

Samsung Biologics Co., Ltd.
Statements of Cash Flows, Continued
For the years ended December 31, 2018 and 2017

(In thousands of won)

	<u>2018</u>	<u>2017</u>
Cash flows from investing activities		
Decrease in short-term financial instruments	₩ 2,730,000,000	2,270,000,000
Decrease in financial assets at fair value through profit	750,000,000	-
Decrease in long-term financial instruments	-	1,000
Decrease in investments in joint ventures	759,460,612	-
Disposal of property, plant and equipment	13,208	-
Increase in financial assets at fair value through profit	(1,290,000,000)	-
Increase in short-term loans	(600,000)	-
Increase in short-term financial instruments	(3,280,000,000)	(1,300,000,000)
Acquisition of property, plant and equipment	(347,851,891)	(494,119,453)
Acquisition of intangible assets	(5,811,483)	(10,682,088)
Acquisition of investments in associates and joint ventures	(16,405,000)	(200,014,156)
Net cash from (used in) investing activities	<u>(701,194,554)</u>	<u>265,185,303</u>
Cash flows from financing activities		
Proceeds from issue of bonds	189,579,060	49,924,480
Proceeds from short-term borrowings	50,000,000	40,000,000
Proceeds from long-term borrowings	153,112,000	90,000,000
Repayment of bonds	(50,000,000)	(170,000,000)
Repayment of current portion of long-term borrowings	(18,750,000)	(236,562,523)
Repayment of long-term borrowings	-	(60,537,587)
Net cash from (used in) financing activities	<u>323,941,060</u>	<u>(287,175,630)</u>
Net increase (decrease) in cash and cash equivalents	<u>(340,466,186)</u>	<u>127,826,000</u>
Cash and cash equivalents at January 1, 2018	358,088,091	230,337,601
Exchange rate effects on foreign currency in cash and cash equivalents	-	(75,510)
Cash and cash equivalents at December 31, 2018	<u>₩ 17,621,905</u>	<u>358,088,091</u>

See accompanying notes to the financial statements.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

1. Description of the Company

Samsung Biologics Co., Ltd. (the "Company") was established on April 22, 2011 under the commercial law of the Republic of Korea and is engaged in Biosimilar manufacturing. The Company's head office is located in 300, Songdobio-daero, Yeonsu-gu, Incheon-si.

The Company was listed on securities market in November 10, 2016. As of December 31, 2018, the Company's share capital is ₩165,413 million.

The shareholders of the Company are as follows:

Shareholder	Shares	Percentage of ownership
Samsung C&T Corporation	28,742,466	43.44%
Samsung Electronics Co., Ltd.	20,836,832	31.49%
Others	16,585,702	25.07%
	66,165,000	100.00%

2. Basis of Accounting

These financial statements have been prepared in accordance with Korean International Financial Reporting Standards ("K-IFRS"), as prescribed in *the Act on External Audits of Corporations in the Republic of Korea*.

The financial statements were authorized for issue by the Board of Directors on January 31, 2019, which will be submitted for approval to the shareholders' meeting to be held on March 22, 2019.

This is the first set of the financial statements where K-IFRS No.1115 has been applied. Changes to significant accounting policies are described in note 3.

(1) Basis of Measurement

The financial statements have been prepared on the historical cost basis, except for the following material items in the statements of financial position:

- ✓ Derivatives financial instruments measured at fair value
- ✓ Liabilities for defined benefit plans recognized at the net of the total present value of defined benefit obligations less the fair value of plan assets

(2) Functional and Presentation Currency

These financial statements are presented in Korean won, which is the Company's functional currency and the currency of the primary economic environment in which the Company operates.

(3) Use of estimates and judgments

The preparation of the financial statements in conformity with K-IFRS requires management to make judgments, estimates and assumptions that affect the application of accounting policies and the reported amounts of assets, liabilities, income and expenses. Actual results may differ from these estimates.

Estimates and underlying assumptions are reviewed on an ongoing basis. Revisions to accounting estimates are recognized in the period prospectively.

2. Basis of Accounting, Continued

(3) Use of estimates and judgments, continued

(i) Critical judgments

Information about critical judgments in applying accounting policies that have the most significant effect on the amounts recognized in the financial statements is included in the following note:

Note 14 : consolidation scope - whether the Company has de facto control over an investee.

(ii) Assumptions and estimation uncertainties

Information about assumptions and estimation uncertainties that have a significant risk of resulting in a material adjustment within the next financial year are included in the following notes: Note 21 : measurement of defined benefit obligations - key actuarial assumptions; Note 28 : recognition of deferred tax assets - use of tax losses; Note 30 : recognition and measurement of provision and contingencies - key assumptions about likelihood and magnitude of an outflow of resources; and Note 14, Note 16, Note 33 : impairment test - key assumptions underlying recoverable amounts.

(iii) Measurement of fair values

A number of the Company's accounting policies and disclosures require the measurement of fair values for both financial and non-financial assets and liabilities. The Company has an established control framework with respect to the measurement of fair values. This includes a valuation team that has overall responsibility for overseeing all significant fair value measurements, including Level 2 fair values, and reports directly to the Chief Financial Officer (CFO).

The valuation team regularly reviews significant unobservable inputs and valuation adjustments. If third party information, such as broker quotes or pricing services, is used to measure fair values, then the valuation team assesses the evidence obtained from the third parties to support the conclusion that such valuations meet the requirements of K-IFRS, including the level in the fair value hierarchy in which such valuations should be classified.

Significant valuation issues are reported to the Company's Audit Committee.

When measuring the fair value of an asset or a liability, the Company uses market observable data as far as possible. Fair values are categorized into different levels in a fair value hierarchy based on the inputs used in the valuation techniques as follows.

- ✓ Level 1: quoted prices (unadjusted) in active markets for identical assets or liabilities.
- ✓ Level 2: inputs other than quoted prices included in Level 1 that are observable for the asset or liability, either directly (i.e. as prices) or indirectly (i.e. derived from prices).
- ✓ Level 3: inputs for the asset or liability that are not based on observable market data (unobservable inputs).

If the inputs used to measure the fair value of an asset or a liability fall into different levels of the fair value hierarchy, then the fair value measurement is categorized in its entirety in the same level of the fair value hierarchy as the lowest level input that is significant to the entire measurement. The Company recognizes transfers between levels of the fair value hierarchy at the end of the reporting period during which the change has occurred.

Further information about the assumptions made in measuring fair values is included in following note:

Note 6 – Financial instruments

3. Changes in accounting policies

Except for the application of the first and amendment standards as applied beginning January 1, 2018, accounting policies set out below have been applied consistently to all periods presented in these financial statements.

3. Changes in accounting policies, Continued

(1) K-IFRS No. 1115, 'Revenue from Contract with Customers'

K-IFRS No.1115 'Revenue from Contracts with Customers' provides the five-step revenue recognition model to determine whether revenue is recognised, the extent of revenue recognition and the timing of revenue recognition. It replaced revenue recognition guidance existed, including K-IFRS No. 1018, 'Revenue', K-IFRS No. 1011, 'Construction Contracts', K-IFRIC No. 2031, 'Revenue – Barter transactions involving advertising services', K-IFRIC No. 2113, 'Customer Loyalty Programs', K-IFRIC No. 2115, 'Agreements for the construction of real estate', and K-IFRIC No. 2118, 'Transfers of assets from customers'

The Company has adopted K-IFRS No.1115 using the cumulative effect method without practical expedients, with the effect of initially applying this standard recognized at the date of initial application (i.e. January 1, 2018). Accordingly, each line item in the financial statements affected in 2018 has been disclosed as adopting this standard, compared with K-IFRS No. 1018 and related interpretations which were effective before the adoption of the new standard.

The following table summarizes the impact, net of tax, of transition to K-IFRS No.1115 on retained earnings at January 1, 2018.

		Amount
<i>(In thousands of won)</i>		
Retained earnings:		
Product sales – single performance obligation	₩	(15,287,783)
Retained earnings – Equity in investee's income		(34,300,468)
Related tax effect		7,613,342
Impact at January 1, 2018	₩	(41,974,909)

The following table summarizes the impacts of adopting K-IFRS No.1115 on the Company's statement of financial position as at December 31, 2018 and its statements of comprehensive income and cash flows for the annual period then ended for each of the line items affected.

Impact on the statement of financial position as of December 31, 2018 is as follows:

		Adjustments of K-IFRS No.1115	Amounts without adoption of K-IFRS No.1115
		As reported	
<i>(In thousands of won)</i>			
Current assets:			
Inventories	₩	302,360,715	11,815,636
Contract assets		19,499,176	(19,499,176)
Non-current assets:			
Contract assets		77,079,462	(77,079,462)
Investments in associates and joint ventures		2,561,802,521	9,968,797
Current liabilities:			
Contract liabilities		24,794,825	(24,794,825)
Other current liabilities		87,695,614	76,197
Non-current liabilities:			
Contract liabilities		76,309,965	(76,309,965)
Deferred tax liabilities		534,542,330	7,376,713
Equity:			
Retained earnings		1,509,718,830	18,857,675

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3. Changes in accounting policies, Continued

(1) K-IFRS No.1115, 'Revenue from Contract with Customers', Continued

Impact on the statement of comprehensive income for annual period ended December 31, 2018 is as follows:
(In thousands of won)

	As reported	Adjustments of K-IFRS No.1115	Amounts without adoption of K-IFRS No.1115
Revenue	₩ 535,805,801	42,750,084	578,555,885
Cost of revenue	(390,645,859)	(41,775,635)	(432,421,494)
Gross profit	145,159,942	974,449	146,134,391
Operating profit	55,678,745	974,449	56,653,194
Finance income	19,320,789	(2,767)	19,318,022
Finance costs	(24,337,217)	6,126	(24,331,091)
Equity loss on investments in associates and joint ventures	(129,337,333)	(16,280,268)	(145,617,601)
Other income	390,399,125	(8,051,401)	382,347,724
Income before income tax	302,975,837	(23,353,863)	279,621,974
Income tax expense (benefit)	78,866,642	(236,629)	78,630,013
Net income for the period	₩ 224,109,195	(23,117,234)	200,991,961

Impact on the statement of cash flows for annual period ended December 31, 2018 is as follows:
(In thousands of won)

	As reported	Adjustments of K-IFRS No.1115	Amounts without adoption of K-IFRS No.1115
Net cash used in operating activities	₩ 36,787,308	-	36,787,308
Cash generated from operating activities	52,790,833	-	52,790,833
Net profit for the period	224,109,195	(23,117,234)	200,991,961
Adjustments for:	(51,544,858)	24,091,682	(27,453,176)
Loss (reversal of loss) on valuation of inventories	(1,932,844)	18,641,061	16,708,217
Impairment loss on contract assets	18,641,061	(18,641,061)	-
Gain on foreign currency translations	(112,646)	2,767	(109,879)
Loss on foreign currency translations	3,291,913	(6,126)	3,285,787
Gain on disposal of investments in associates and joint ventures	(389,213,672)	8,051,401	(381,162,271)
Equity loss on investments in associates and joint ventures	129,337,333	16,280,268	145,617,601
Income tax expense (benefit)	78,866,642	(236,629)	78,630,013
Changes in assets and liabilities:	(119,773,504)	(974,448)	(120,747,952)
Decrease (increase) in inventories	(90,441,155)	(30,456,698)	(120,897,853)
Decrease (increase) in contract assets	(73,070,459)	73,070,459	-
Increase (decrease) in contract liabilities	43,664,407	(43,664,407)	-
Increase (decrease) in other current liabilities	31,209,439	76,197	31,285,636

The details of the new significant accounting policies and the nature of the changes to previous accounting policies in relation to the Company's goods and services are set out below. Under K-IFRS No.1115, revenue is recognized when a customer obtains control of the goods or services. Determining the timing of the transfer of control – at a point in time or over time – requires judgment.

3. Changes in accounting policies, Continued

(1) K-IFRS No.1115, 'Revenue from Contract with Customers', Continued

Type of product/service	Nature, timing of satisfaction of performance obligations, significant payment terms	Nature of change in accounting policy
CMO business	The Company manufactures and supplies made-to-order biopharmaceutical products. Goods manufactured do not have an alternative use to the Company, and it has an enforceable right to payment for performance completed to date. This is because that the Company produces biomedicine at a quality that a customer requires and is entitled to reimbursement of the costs incurred to date, including a reasonable margin if a customer terminates a contract. Therefore, the Company recognizes revenue and the associated costs over time as it satisfies its performance obligation over a contract period. Services mandatorily rendered prior to a production of biomedicine do not form a part of a performance obligation since they are considered as a set-up process.	In case of sale of biopharmaceuticals, the Company recognized revenue at a point of customer's quality release after a completion of production as significant risks and rewards of ownership of goods are considered to be transferred to a customer at such point under K-IFRS 1018. Revenue from services rendered is recognized in profit when they are transferred to a customer under K-IFRS 1018. In accordance with a new accounting policy, only sale of biopharmaceuticals is considered to be a performance obligation; therefore, the Company allocates and recognizes revenue over time based on a point when control of biopharmaceuticals is transferred to a customer. As a result of such change, contract assets and contract liabilities are newly recognized.
CDO business	The Company provides cell line and upstream/downstream process development services. As each research and development stages is linked together and is highly interdependent and highly interrelated, the Company determines that a contract includes a single performance obligation. Also, revenue is recognized over time during a contract period as a customer simultaneously receives and consumes the benefits provided by the Company's performance as the Company performs.	Revenue is recognized based on consideration allocated over the period of each research and development stage under K-IFRS 1018. In accordance with a new accounting policy, all stages of research and development are considered to be a single performance obligation, and total contract price is recognized as revenue based on measure of progress. As a result of such change, contract assets and contract liabilities are newly recognized.

(2) K-IFRIC No. 2122, 'Foreign Currency Transactions and Advance Consideration'

According to the enacted Interpretation, the date of transaction for the purpose of determining the exchange rate to use on initial recognition of the related asset, expense or income (or part of it) is the date on which an entity initially recognizes the non-monetary asset or non-monetary liability arising from the prepayment or receipt of advance consideration. In case there are multiple payments or receipts in advance, the entity should determine a date of the transaction for each payment or receipt of advance consideration. The impact of the amendments on the financial statements is not significant.

(3) New standards and interpretations not yet adopted

The following new standards, interpretations, and amendments to existing standards have been published and are mandatory for the Company for annual periods beginning on January 1, 2018, and the Company has not early adopted them.

3. Changes in accounting policies, Continued

(3) New standards and interpretations not yet adopted, Continued

To assess the financial impact of the initial application of K-IFRS No.1116, this will have a significant effect on the financial statements and the Company have completed the specific impact on financial statements.

1) K-IFRS No.1116 Leases

K-IFRS No.1116 replaces existing leases guidance, including K-IFRS No.1017 Leases, K-IFRIC No.2104 Determining whether an Arrangement contains a Lease, K-IFRS No.2015 Operating Leases – Incentives and K-IFRS No.2027 Evaluating the Substance of Transactions Involving the Legal Form of a Lease.

The Company is required to adopt K-IFRS No.1116 Leases from 1 January 2019.

- Lessee accounting - User Model

K-IFRS No. 1116 suggests a single accounting model that requires a lessee to recognize lease related asset and liability in the financial statements. A lessee is required to recognize a right-of-use asset representing its right to use the underlying leased asset and a lease liability representing its obligation to make lease payments. The lessee may elect not to apply the requirements to short-term lease (a lease term of 12 months or less at the commencement date) or less at the commencement date and low value assets (e.g. underlying assets below \$5,000).

- Lessor accounting – Risk and rewards model

Accounting treatment for lessor is similar to the existing standard which classifies lease into finance and operating lease.

When applying K-IFRS No. 1116, the Company does not expect the enactments to have a significant impact on the financial statements as the accounting standard as a lessor did not change significantly from K-IFRS No. 1017.

2) Lessee accounting

A lessee shall apply this standard to its leases either: a) retrospectively to each prior reporting period presented applying K-IFRS No. 1008, Changes in Accounting Estimates and Errors (Full retrospective application) or b) retrospectively with the cumulative effect of initially applying the standard recognized at the date of initial application.

At the time of contract agreement, our Company determines whether the contract itself is a lease or includes a lease, and also at the date of initial application, the Company identifies whether the contract is a lease or includes a lease in accordance with the standards. However, exceptions will be applied to short-term leases and leases related to small assets from the standards.

The Company plans to apply K-IFRS No. 1116 retrospectively with the cumulative effect of initially applying the standard as at January 1, 2019. Accordingly, the right-of-use assets measured at the date of initial application will be the same amount as the lease liabilities, and the amount of advance or accrued lease payments will be adjusted in relation to the lease recognized in the financial statements immediately before the date of initial application. In addition, the cumulative effect of applying K-IFRS NO.1116 is adjusted for retained earnings at the date of initial application. The Company will not restate any comparative information.

3. Changes in accounting policies, Continued

(3) New standards and interpretations not yet adopted, Continued

3) Impact assessment to identify financial effects

The Company performed an impact assessment to identify potential financial effects of applying K-IFRS No. 1116. The assessment was performed based on available information as at December 31, 2018 to identify effects on 2019 financial statements.

The total minimum lease payment expected to be paid by the Company in relation to operating leases before discount to present value is ₩ 9,471 million as at January 1, 2019. When the payment is discounted at the incremental borrowing rate of the lessee, the total minimum lease payment amounts to ₩ 8,969 million.

For contracts that are, or contain, a lease, the Company plans to apply the practical expedient to account for each lease component and any associated non-lease components as a single lease component.

As a result of applying the standard, the Company expects the underlying lease assets and lease liabilities as at January 1, 2019 to increase by ₩ 9,912 million and ₩ 8,969 million, respectively.

The results of the assessment may change due to additional information that the Company may obtain subsequent to performing the assessment.

4) Lessor accounting

As a lessor, the Company can alter accounting procedure of classification of lease contracts and separation of lease components in accordance with K-IFRS No 1116, and as an intermediate lease provider, the Company can reassess the classification of operating leases and finance leases for all subleases classified as operating leases before the date of initial application and maintained at the date of initial application in accordance with K-IFRS No 1116. The Company performed an impact assessment to identify potential financial effects of applying K-IFRS No. 1116. And the Company does not expect the enactments to have a significant impact on the financial statements. The results of the assessment may change due to additional information that the Company may obtain subsequent to performing the assessment.

4. Significant Accounting Policies

The significant accounting policies applied by the Company in preparation of its financial statements in accordance with K-IFRS are included below. The accounting policies set out below have been applied consistently to all periods presented in these financial statements except for the changes in accounting policies as explained in Note 3.

(1) Interests in equity-accounted investees

The Company's interests in equity-accounted investees comprise interests in associates and joint ventures.

Associates are those entities in which the company has significant influence, but no control or joint control, over the financial and operating policies. A joint venture is an arrangement in which the Company has joint control, whereby the Company has rights to the net assets of the arrangement, rather than rights to its assets and obligations for its liabilities.

4. Significant Accounting Policies, Continued

(1) Interests in equity-accounted investees, Continued

Interests in associates and the joint venture are accounted for using the equity method. They are initially recognized at cost, which includes transaction costs. Subsequent to initial recognition, the financial statements include the Company's share of the profit or loss and OCI of equity-accounted investees until the date on which significant influence or joint control ceases.

(2) Cash and cash equivalents

Cash and cash equivalents comprise cash balances and call deposits with maturities of three months or less from the acquisition date that are subject to an insignificant risk of changes in their fair value and are used by the Company in the management of its short-term commitments. Equity investments are excluded from cash equivalents unless they are, in substance, cash equivalents, for example in the case of preferred shares when it has a short maturity with a specified redemption date.

(3) Inventories

The cost of inventories is based on the moving-weighted average principle, except for goods in transit based on specific identification method and includes expenditures for acquiring the inventories, production or conversion costs and other costs incurred in bringing them to their existing location and condition. The allocation of fixed production overheads to the costs of finished goods or work in progress are based on the normal capacity of the production facilities.

Inventories are measured at the lower of cost and net realizable value. The amount of any write-down of inventories to net realizable value and all losses of inventories are recognized as an expense in the period the write-down or loss occurs. The amount of any reversal of any write-down of inventories, arising from an increase in net realizable value, are recognized as a reduction in the amount of inventories recognized as an expense in the period in which the reversal occurs.

(4) Property, plant and equipment

Property, plant and equipment are initially measured at cost. The cost of property, plant and equipment includes expenditures arising directly from the construction or acquisition of the asset, any costs directly attributable to bringing the asset to the location and condition necessary for it to be capable of operating in the manner intended by management, and the initial estimate of the costs of dismantling and removing the item and restoring the site on which it is located.

Subsequent to initial recognition, an item of property, plant and equipment shall be carried at its cost less any accumulated depreciation and any accumulated impairment losses.

Property, plant and equipment, except for land, are depreciated over estimated useful lives that appropriately reflect the pattern in which the asset's future economic benefits are expected to be consumed on a straight-line basis.

Gains and losses on disposal of an item of property, plant and equipment are determined by comparing the proceeds from disposal with the carrying amount of property, plant and equipment and are recognized in profit or loss.

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4. Significant Accounting Policies, Continued

(4) Property, plant and equipment, Continued

The estimated useful lives of the property, plant and equipment are as follows:

	Useful lives (years)
Buildings and structures	40
Machinery	5~15
Vehicles	4
Other tangible assets	4

Depreciation methods, useful lives and residual values are reviewed at the end of each reporting date and adjusted, if appropriate. The change is accounted for as a change in an accounting estimate.

(5) Intangible assets

Intangible assets are measured initially at cost and, subsequently, are carried at cost less accumulated amortization and accumulated impairment losses.

Amortization of intangible assets is calculated on a straight-line basis over the following estimated useful lives of intangible assets from the date that they are available for use. The residual value of intangible assets is zero. However, as there are no foreseeable limits to the periods over which memberships are expected to be available for use, they are determined as having indefinite useful lives and not amortized.

	Useful lives (years)
Software	5
Other intangible assets	5
Land usage rights	50
Memberships	indefinite

(6) Borrowing costs

The Company capitalizes borrowing costs directly attributable to the acquisition, construction, or production of a qualifying asset as part of the cost of that asset. Other borrowing costs are recognized in expense as incurred. A qualifying asset is an asset that requires a substantial period of time to get ready for its intended use or sale. Financial assets and inventories that are manufactured or otherwise produced over a short period of time are not qualifying assets. Assets that are ready for their intended use or sale when acquired are not qualifying assets.

To the extent that the Company borrows funds specifically for the purpose of obtaining a qualifying asset, the Company determines the amount of borrowing costs eligible for capitalization as the actual borrowing costs incurred on that borrowing during the period less any investment income on the temporary investment of those borrowings. To the extent that the Company borrows funds generally and uses them for the purpose of obtaining a qualifying asset, the Company shall determine the amount of borrowing costs eligible for capitalization by applying a capitalization rate to the expenditures on that asset. The capitalization rate shall be the weighted average of the borrowing costs applicable to the borrowings of the Company that are outstanding during the period, other than borrowings made specifically for the purpose of obtaining a qualifying asset. The amount of borrowing costs that the Company capitalizes during a period shall not exceed the amount of borrowing costs incurred during that period.

4. Significant Accounting Policies, Continued

(7) Government grants

Government grants are not recognized unless there is reasonable assurance that the Company will comply with the grant's conditions and that the grant will be received.

Government grants whose primary condition is that the Company purchases, constructs or otherwise acquires long-term assets are deducted in calculating the carrying amount of the asset. The grant is recognized in profit or loss over the life of a depreciable asset as a reduction to depreciation expense.

(8) Impairment of non-financial assets

The carrying amounts of the Company's non-financial assets, except assets arising from employee benefits, inventories and deferred tax assets, are reviewed at the end of the reporting period to determine whether there is any indication of impairment. If any such indication exists, then the asset's recoverable amount is estimated. Goodwill and intangible assets that have indefinite useful lives or that are not yet available for use, irrespective of whether there is any indication of impairment, are tested for impairment annually by comparing their recoverable amount to their carrying amount.

For impairment testing, assets are grouped together into the smallest group of assets that generates cash inflows that are largely independent of the cash inflows from other assets or groups of assets ("CGUs"). The recoverable amount of an asset or CGU is the greater of its value in use and its fair value less costs to sell. The value in use is estimated by applying a pre-tax discount rate that reflect current market assessments of the time value of money and the risks specific to the asset or CGU for which estimated future cash flows have not been adjusted, to the estimated future cash flows expected to be generated by the asset or CGU.

An impairment loss is recognized in profit or loss if the carrying amount of an asset or a CGU exceeds its recoverable amount.

(9) Leases

The Company classifies and accounts for leases as either a finance or operating lease, depending on the terms. Leases where the Company assumes substantially all of the risks and rewards of ownership are classified as finance leases. All other leases are classified as operating leases.

① Finance leases

At the commencement of the lease term, the Company recognizes as finance assets and finance liabilities in its separate statements of financial position, the lower amount of the fair value of the leased property and the present value of the minimum lease payments, each determined at the inception of the lease. Any initial direct costs are added to the amount recognized as an asset.

Minimum lease payments are apportioned between the finance charge and the reduction of the outstanding liability. The finance charge is allocated to each period during the lease term so as to produce a constant periodic rate of interest on the remaining balance of the liability. Contingent rents are charged as expenses in the periods in which they are incurred.

The depreciable amount of a leased asset is allocated to each accounting period during the period of expected use on a systematic basis consistent with the depreciation policy the lessee adopts for depreciable assets that are owned. If there is no reasonable certainty that the lessee will obtain ownership by the end of the lease term, the asset is fully depreciated over the shorter of the lease term and its useful life. The Company reviews to determine whether the leased asset may be impaired.

4. Significant Accounting Policies, Continued

(9) Leases, Continued

② Operating leases

Leases where the lessor retains a significant portion of the risks and rewards of ownership are classified as operating leases. Payments made under operating leases (net of any incentives received from the lessor) are recognized in profit or loss on a straight-line basis over the period of the lease.

③ Determining whether an arrangement contains a lease

Determining whether an arrangement is, or contains, a lease shall be based on the substance of the arrangement and requires an assessment of whether fulfillment of the arrangement is dependent on the use of a specific asset or assets (the asset) and the arrangement conveys a right to use the asset.

At inception or reassessment of the arrangement, the Company separates payments and other consideration required by such an arrangement into those for the lease and those for other elements on the basis of their relative fair values. If the Company concludes for a financial lease that it is impracticable to separate the payments reliably, the Company recognizes an asset and a liability at an amount equal to the fair value of the underlying asset that was identified as the subject of the lease. Subsequently, the liability shall be reduced as payments are made and an imputed finance charge on the liability recognized using the purchaser's incremental borrowing rate of interest.

(10) Employee benefits

① Short-term employee benefits

Short-term employee benefits are employee benefits that are expected to be settled wholly before twelve months after the end of the annual reporting period in which the employees render the related services. When an employee has rendered service to the Company during an accounting period, the Company recognizes the undiscounted amount of short-term employee benefits expected to be paid in exchange for that service.

② Other long-term employee benefits

Other long-term employee benefits include employee benefits that are settled beyond 12 months after the end of the period in which the employees render the related service, and are calculated at the present value of the amount of future benefit that employees have earned in return for their service in the current and prior periods. Any changes from re-measurements are recognized through profit or loss in the period in which they arise.

③ Retirement benefits: defined benefit plans

The Company's net obligation in respect of defined benefit plans is calculated separately for each plan by estimating the amount of future benefit that employees have earned in the current and prior periods, discounting that amount and deducting the fair value of any plan assets.

The calculation of defined benefit obligations is performed annually by a qualified actuary using the projected unit credit method. When the calculation results in a potential asset for the Company, the recognized asset is limited to the present value of economic benefits available in the form of any future refunds from the plan or reductions in future contributions to the plan. To calculate the present value of economic benefits, consideration is given to any applicable minimum funding requirements.

4. Significant Accounting Policies, Continued

(10) Employee benefits, Continued

Remeasurements of the net defined benefit liability, which comprise actuarial gains and losses, the return on plan assets (excluding interest), and the effect of the asset ceiling (if any, excluding interest), are recognized immediately in OCI. The Company determines the net interest expense (income) on the net defined benefit liability (asset) for the period by applying the discount rate used to measure the defined benefit obligation at the beginning of the annual period to the then-net defined benefit liability (asset), taking into account any changes in the net defined benefit liability (asset) during the period as a result of contributions and benefit payments. Net interest expense and other expenses related to defined benefit plans are recognized in profit or loss.

When the plan amendment or curtailment occurs, gains or losses on amendment or curtailment in benefits for the past service provided are recognized through profit or loss. The Company recognizes gain or loss on a settlement when the settlement of defined benefit plan occurs.

(11) Provisions

Provisions are recognized when the Company has a present legal or constructive obligation as a result of a past event, it is probable that an outflow of resources embodying economic benefits will be required to settle the obligation and a reliable estimate can be made of the amount of the obligation.

The risks and uncertainties that inevitably surround many events and circumstances are taken into account in reaching the best estimate of a provision. Where the effect of the time value of money is material, provisions are determined at the present value of the expected future cash flows.

Where some or all of the expenditures required to settle a provision are expected to be reimbursed by another party, the reimbursement is recognized when, and only when, it is virtually certain that reimbursement will be received if the entity settles the obligation. The reimbursement is presented as a separate asset.

Provisions are reviewed at the end of each reporting period and adjusted to reflect the current best estimates. If it is no longer probable that an outflow of resources embodying economic benefits will be required to settle the obligation, the provision is reversed.

A provision is used only for expenditures for which the provision was originally recognized.

4. Significant Accounting Policies, Continued

(12) Emissions rights

The Company accounts for greenhouse gases emission right and the relevant liability as below pursuant to the *Act on Allocation and Trading of Greenhouse Gas Emission*.

① Greenhouse gases emission right

Greenhouse gases emission right consists of emission allowances which are allocated from the government free of charge or purchased from the market. The cost includes any directly attributable costs incurred during the normal course of business.

Emission rights held for the purpose of performing the obligation is classified as intangible asset and is initially measured at cost. After initial recognition, they are carried at cost less accumulated impairment losses. Emission rights held for short-swing profits are classified as current asset and are measured at fair value with any changes in fair value recognized as profit or loss in the respective reporting period.

The Company derecognizes an emission right asset when the emission allowance is unusable, disposed, or submitted to government in which the future economic benefits are no longer expected to be probable.

② Emission liability

Emission liability is a present obligation of submitting emission rights to the government with regard to emission of greenhouse gas. Emission liability is recognized when it is probable that outflows of resources will be required to settle the obligation and the costs required to perform the obligation are reliably estimable. Emission liability is an amount of estimated obligations for emission rights to be submitted to the government for the performing period. The emission liability is measured based on the expected quantity of emission for the performing period in excess of emission allowance in possession and the unit price for such emission rights in the market at the end of the reporting period.

(13) Foreign currency transactions

Transactions in foreign currencies are translated to the respective functional currency of the Company at exchange rates at the dates of the transactions. Monetary assets and liabilities denominated in foreign currencies are retranslated to the functional currency using the reporting date's exchange rate. Non-monetary assets and liabilities denominated in foreign currencies that are measured at fair value are retranslated to the functional currency at the exchange rate at the date that the fair value was determined.

Foreign currency differences arising on retranslation are recognized in profit or loss, except for differences arising on the retranslation of available-for-sale equity instruments, a financial liability designated as a hedge of the net investment in a foreign operation, or qualifying cash flow hedges, which are recognized in other comprehensive income.

(14) Equity capital

Ordinary shares are classified as equity. Incremental costs directly attributable to the issuance of ordinary shares and share options are recognized as a deduction from equity, net of any tax effects.

(15) Revenue from Contracts with Customers

The Company has initially applied K-IFRS No.1115 "Revenue from Contracts with Customers" from January 1, 2018. The company's accounting policies related to "Revenue from Contracts with Customers" and the initial application effect of K-IFRS No.1115 are described in Note 3(1).

4. Significant Accounting Policies, Continued

(16) Finance income and finance costs

Finance income comprises interest income on funds invested. Interest income is recognized as it accrues in profit or loss, using the effective interest method.

Finance costs comprise interest expense on borrowings and changes in the fair value of derivatives financial instruments at fair value through profit or loss. Borrowing costs that are not directly attributable to the acquisition, construction, or production of a qualifying asset are recognized in profit or loss using the effective interest method.

(17) Income taxes

Income tax expense comprises current and deferred tax. Current tax and deferred tax are recognized in profit or loss except to the extent that it relates to a business combination or items recognized directly in equity or in other comprehensive income.

If the Company determines that interests and penalties from income taxes fall under income taxes, it accounts them for in accordance with K-IFRS No.1012 'Income taxes'. If not, then it applies K-IFRS No.1037 'Provisions, Contingent Liabilities and Contingent Assets'.

① Current tax

Current tax is the expected tax payable or receivable on the taxable profit or loss for the year, using tax rates enacted or substantively enacted at the end of the reporting period and any adjustment to tax payable in respect of previous years. The taxable profit is different from the accounting profit for the period since the taxable profit is calculated excluding the temporary differences, which will be taxable or deductible in determining taxable profit (tax loss) of future periods, and non-taxable or non-deductible items from the accounting profit.

② Deferred tax

Deferred tax is recognized, using the asset-liability method, in respect of temporary differences between the carrying amounts of assets and liabilities for financial reporting purposes and the amounts used for taxation purposes. A deferred tax liability is recognized for all taxable temporary differences. A deferred tax asset is recognized for all deductible temporary differences to the extent that it is probable that taxable profit will be available against which they can be utilized. However, deferred tax is not recognized for taxable temporary differences arising on the initial recognition of goodwill, or the initial recognition of assets or liabilities in a transaction that is not a business combination and that affects neither accounting profit or loss nor taxable income.

If future taxable profit is likely to occur, unused tax losses, tax credits, and deductible temporary differences are recognized as deferred tax assets within a range of future taxable profit. Future taxable profit is determined from the expiration of related taxable temporary differences. If taxable temporary difference is not sufficient to fully recognize the deferred tax asset, the expiration of temporary differences will be considered.

The measurement of deferred tax liabilities and deferred tax assets reflects the tax consequences that would follow from the manner in which the Company expects, at the end of the reporting period, to recover or settle the carrying amount of its assets and liabilities.

The Company recognizes a deferred tax liability for all taxable temporary differences associated with investments in subsidiaries, associates, and interests in joint ventures, except to the extent that the Company is able to control the timing of the reversal of the temporary difference and it is probable that the temporary difference will not reverse in the foreseeable future. The Company recognizes a deferred tax asset for all deductible temporary differences arising from investments in subsidiaries and associates, to the extent that it is probable that the temporary difference will reverse in the foreseeable future and taxable profit will be available against which the temporary difference can be utilized.

4. Significant Accounting Policies, Continued

(17) Income taxes, Continued

The carrying amount of a deferred tax asset is reviewed at the end of each reporting period and reduced to the extent that it is no longer probable that sufficient taxable profit will be available to allow the benefit of part or all of that deferred tax asset to be utilized.

Deferred tax assets and liabilities are measured at the tax rates that are expected to apply to the period when the asset is realized or the liability is settled, based on tax rates (and tax laws) that have been enacted or substantively enacted by the end of the reporting period.

Deferred tax assets and liabilities are offset only if there is a legally enforceable right to the Company to offset the related current tax liabilities and assets, they relate to income taxes levied by the same tax authority, and the Company intends to settle current tax liabilities and assets on a net basis.

(18) Earnings per share

The Company presents basic and diluted earnings per share (EPS) data for its ordinary shares. Basic EPS is calculated by dividing the profit or loss attributable to ordinary shareholders of the Company by the weighted average number of ordinary shares outstanding during the period, adjusted for own shares held. The Company has no dilutive potential shares, therefore diluted loss per share is equal to the basic earnings (loss) per share.

(19) Non-derivative financial assets

① Recognition and initial measurement

Trade receivables and debt securities issued are initially recognised when they are originated. All other financial assets and financial liabilities are initially recognised when the Company becomes a party to the contractual provisions of the instrument.

A financial asset (unless it is a trade receivable without a significant financing component) or financial liability is initially measured at fair value plus, for an item not at FVTPL, transaction costs that are directly attributable to its acquisition or issue. A trade receivable without a significant financing component is initially measured at the transaction price.

② Classification and subsequent measurement

On initial recognition, a financial asset is classified as measured at: amortised cost; FVOCI – debt investment; FVOCI – equity investment; or FVTPL.

Financial assets are not reclassified subsequent to their initial recognition unless the Company changes its business model for managing financial assets, in which case all affected financial assets are reclassified on the first day of the first reporting period following the change in the business model.

A financial asset is measured at amortised cost if it meets both of the following conditions and is not designated as at FVTPL:

- it is held within a business model whose objective is to hold assets to collect contractual cash flows; and
- its contractual terms give rise on specified dates to cash flows that are solely payments of principal and interest on the principal amount outstanding.

A debt investment is measured at FVOCI if it meets both of the following conditions and is not designated as at FVTPL:

- it is held within a business model whose objective is achieved by both collecting contractual cash flows and selling financial assets; and

4. Significant Accounting Policies, Continued

(19) Non-derivative financial assets, Continued

- its contractual terms give rise on specified dates to cash flows that are solely payments of principal and interest on the principal amount outstanding.

On initial recognition of an equity investment that is not held for trading, the Company may irrevocably elect to present subsequent changes in the investment's fair value in OCI. This election is made on an investment-by-investment basis.

All financial assets not classified as measured at amortised cost or FVOCI as described above are measured at FVTPL. On initial recognition, the Company may irrevocably designate a financial asset that otherwise meets the requirements to be measured at amortised cost or at FVOCI as at FVTPL if doing so eliminates or significantly reduces an accounting mismatch that would otherwise arise. However, it is impossible to cancel after designating.

1) Financial assets – Business model

The Company makes an assessment of the objective of the business model in which a financial asset is held at a portfolio level because this best reflects the way the business is managed and information is provided to management. The information considered includes:

- the stated policies and objectives for the portfolio and the operation of those policies in practice. These include whether management's strategy focuses on earning contractual interest income, maintaining a particular interest rate profile, matching the duration of the financial assets to the duration of any related liabilities or expected cash outflows or realizing cash flows through the sale of the assets;
- how the performance of the portfolio is evaluated and reported to the Company's management;
- the risks that affect the performance of the business model (and the financial assets held within that business model) and how those risks are managed;
- how managers of the business are compensated – e.g. whether compensation is based on the fair value of the assets managed or the contractual cash flows collected; and
- the frequency, volume and timing of sales of financial assets in prior periods, the reasons for such sales and expectations about future sales activity.

Transfers of financial assets to third parties in transactions that do not qualify for derecognition are not considered sales for this purpose, consistent with the Company's continuing recognition of the assets. Financial assets that are held for trading or are managed and whose performance is evaluated on a fair value basis are measured at FVTPL.

2) Financial assets – Assessment whether contractual cash flows are solely payments of principal and interest

For the purposes of this assessment, 'principal' is defined as the fair value of the financial asset on initial recognition. 'Interest' is defined as consideration for the time value of money and for the credit risk associated with the principal amount outstanding during a particular period of time and for other basic lending risks and costs (e.g. liquidity risk and administrative costs), as well as a profit margin.

In assessing whether the contractual cash flows are solely payments of principal and interest, the Company considers the contractual terms of the instrument. This includes assessing whether the financial asset contains a contractual term that could change the timing or amount of contractual cash flows such that it would not meet this condition.

In making this assessment, the Company considers:

- contingent events that would change the amount or timing of cash flows;
- terms that may adjust the contractual coupon rate, including variable-rate features;
- prepayment and extension features; and
- terms that limit the Company's claim to cash flows from specified assets (e.g. non-recourse features).

4. Significant Accounting Policies, Continued

(19) Non-derivative financial assets, Continued

A prepayment feature is consistent with the solely payments of principal and interest criterion if the prepayment amount substantially represents unpaid amounts of principal and interest on the principal amount outstanding, which may include reasonable additional compensation for early termination of the contract.

Additionally, for a financial asset acquired at a discount or premium to its contractual per amount, a feature that permits or requires prepayment at an amount that substantially represents the contractual par amount plus accrued (but unpaid) contractual interest (which may also include reasonable addition a compensation for early termination) is treated as consistent with this criterion if the fair value of the prepayment feature is insignificant at initial recognition.

3) Financial assets – Subsequent measurement and gains and losses

Financial assets at FVTPL	These assets are subsequently measured at fair value. Net gains and losses, including any interest or dividend income, are recognized in profit or loss.
Financial assets at amortized cost	These assets are subsequently measured at amortized cost using the effective interest method. The amortized cost is reduced by impairment losses. Interest income, foreign exchange gains and losses and impairment are recognized in profit or loss. Any gain or loss on derecognition is recognized in profit or loss.
Debt investments at FVOCI	These assets are subsequently measured at fair value. Interest income calculated using the effective interest method, foreign exchange gains and losses and impairment are recognized in profit or loss. Other net gains and losses are recognized in OCI. On derecognition, gains and losses accumulated in OCI are reclassified to profit or loss.
Equity investments at FVOCI	These assets are subsequently measured at fair value. Dividends are recognized as income in profit or loss unless the dividend clearly represents a recovery of part of the cost of the investment. Other net gains and losses are recognized in OCI and are never reclassified to profit or loss.

③ De-recognition

The Company derecognizes a financial asset when the contractual rights to the cash flows from the financial asset expire, or it transfers the rights to receive the contractual cash flows in a transaction in which substantially all of the risks and rewards of ownership of the financial asset are transferred or in which the Company neither transfers nor retains substantially all of the risks and rewards of ownership and it does not retain control of the financial asset.

The Company enters into transactions whereby it transfers assets recognized in its statement of financial position, but retains either all or substantially all of the risks and rewards of the transferred assets. In these cases, the transferred assets are not derecognized.

④ Offsetting between financial assets and financial liabilities

Financial assets and financial liabilities are offset and the net amount presented in the statement of financial position when, and only when, the Company currently has a legally enforceable right to set off the amounts and it intends either to settle them on a net basis or to realise the asset and settle the liability simultaneously.

(20) Non-derivative financial liabilities

① Financial liabilities at fair value through profit or loss

Financial liabilities are classified as held for trading if they are classified as held for trading, are derivatives, or are initially recognized at fair value through profit or loss. Subsequent to initial recognition, financial liabilities at fair value through profit or loss are measured at fair value, and changes therein are recognized in profit or loss. Upon initial recognition, transaction costs that are directly attributable to the acquisition are recognized in profit or loss as incurred.

4. Significant Accounting Policies, Continued

(20) Non-derivative financial liabilities, Continued

② Other financial liabilities

Non-derivative financial liabilities other than financial liabilities at fair value through profit or loss are classified as other financial liabilities. At the date of initial recognition, other financial liabilities are measured at fair value minus transaction costs that are directly attributable to the acquisition. Subsequent to initial recognition, other financial liabilities are measured at amortized cost using the effective interest method.

③ De-Recognition of financial liabilities

The Company derecognizes a financial liability from the consolidated statement of financial position when it is extinguished (i.e. when the obligation specified in the contract is discharged, cancelled or expires). If the contractual terms of the financial liabilities have changed substantially, the Company de-recognizes the existing liabilities and recognizes the new financial liabilities at fair value on the basis of the new contract.

When a financial liability is de-recognized, the difference between the carrying amount and the consideration paid (including transferred non-cash assets or liabilities assumed) is recognized in profit or loss.

(21) Derivative financial instruments, and hedge accounting

① Derivative financial instruments, and hedge accounting

The Company holds derivative financial instruments to hedge its foreign currency and interest rate risk exposures. Embedded derivatives are separated from the host contract and accounted for separately if the host contract is not a financial asset and certain criteria are met.

Derivatives are initially measured at fair value. Subsequent to initial recognition, derivatives are measured at fair value, and changes therein are generally recognized in profit or loss.

The Company designates certain derivatives as hedging instruments to hedge the variability in cash flows associated with highly probable forecast transactions arising from changes in foreign exchange rates and interest rates and certain derivatives and non-derivative financial liabilities as hedges of foreign exchange risk on a net investment in a foreign operation.

At inception of designated hedging relationships, the Company documents the risk management objective and strategy for undertaking the hedge. The Company also documents the economic relationship between the hedged item and the hedging instrument, including whether the changes in cash flows of the hedged item and hedging instrument are expected to offset each other.

② Cash flow hedges

When a derivative is designated as a cash flow hedging instrument, the effective portion of changes in the fair value of the derivative is recognised in OCI and accumulated in the hedging reserve. The effective portion of changes in the fair value of the derivative that is recognised in OCI is limited to the cumulative change in fair value of the hedged item, determined on a present value basis, from inception of the hedge. Any ineffective portion of changes in the fair value of the derivative is recognised immediately in profit or loss.

The Company designates only the change in fair value of the spot element of forward exchange contracts as the hedging instrument in cash flow hedging relationships. The change in fair value of the forward element of forward exchange contracts ('forward points') is separately accounted for as a cost of hedging and recognised in a costs of hedging reserve within equity.

4. Significant Accounting Policies, Continued

(21) Derivative financial instruments, and hedge accounting, Continued

When the hedged forecast transaction subsequently results in the recognition of a non-financial item such as inventory, the amount accumulated in the hedging reserve and the cost of hedging reserve is included directly in the initial cost of the non-financial item when it is recognised.

For all other hedged forecast transactions, the amount accumulated in the hedging reserve and the cost of hedging reserve is reclassified to profit or loss in the same period or periods during which the hedged expected future cash flows affect profit or loss.

If the hedge no longer meets the criteria for hedge accounting or the hedging instrument is sold, expires, is terminated or is exercised, then hedge accounting is discontinued prospectively. When hedge accounting for cash flow hedges is discontinued, the amount that has been accumulated in the hedging reserve remains in equity until, for a hedge of a transaction resulting in the recognition of a non-financial item, it is included in the non-financial item's cost on its initial recognition or, for other cash flow hedges, it is reclassified to profit or loss in the same period or periods as the hedged expected future cash flows affect profit or loss.

If the hedged future cash flows are no longer expected to occur, then the amounts that have been accumulated in the hedging reserve and the cost of hedging reserve are immediately reclassified to profit or loss.

③ Net investment hedges

When a derivative instrument or a non-derivative financial liability is designated as the hedging instrument in a hedge of a net investment in a foreign operation, the effective portion of, for a derivative, changes in the fair value of the hedging instrument or, for a non-derivative, foreign exchange gains and losses is recognised in OCI and presented in the translation reserve within equity. Any ineffective portion of the changes in the fair value of the derivative or foreign exchange gains and losses on the non-derivative is recognised immediately in profit or loss.

The amount recognised in OCI is reclassified to profit or loss as a reclassification adjustment on disposal of the foreign operation

(22) Impairment of Non-derivative financial assets

① Non-derivative financial instruments and contract assets

The Company recognizes loss allowances for ECLs on:

- financial assets measured at amortized cost;
- debt investments measured at FVOCI; and contract assets.
- Contract assets defined in K-IFRS No.1115

The Company measures loss allowances at an amount equal to lifetime ECLs, except for the following, which are measured at 12-month ECLs:

- debt securities that are determined to have low credit risk at the reporting date;
- other debt securities and bank balances for which credit risk (i.e. the risk of default occurring over the expected life of the financial instrument) has not increased significantly since initial recognition.

Loss allowances for trade receivables and contract assets are always measured at an amount equal to lifetime ECLs.

When determining whether the credit risk of a financial asset has increased significantly since initial recognition and when estimating ECLs, the Company considers reasonable and supportable information that is relevant and available without undue cost or effort. This includes both quantitative and qualitative information and analysis, based on the Company's historical experience and informed credit assessment and including forward-looking information.

4. Significant Accounting Policies, Continued

(22) Impairment of Non-derivative financial assets, Continued

The Company assumes that the credit risk on a financial asset has increased significantly if it is more than 30 days past due.

The Company considers a financial asset to be in default when:

- the borrower is unlikely to pay its credit obligations to the Company in full, without recourse by the Company to actions such as realizing security (if any is held);
- the financial asset is more than 90 days past due.

Lifetime ECLs are the ECLs that result from all possible default events over the expected life of a financial instrument. 12-month ECLs are the portion of ECLs that result from default events that are possible within the 12 months after the reporting date (or a shorter period if the expected life of the instrument is less than 12 months).

The maximum period considered when estimating ECLs is the maximum contractual period over which the Company is exposed to credit risk. months).

② Measurement of ECLs

ECLs are a probability-weighted estimate of credit losses. Credit losses are measured as the present value of all cash shortfalls (i.e. the difference between the cash flows due to the entity in accordance with the contract and the cash flows that the Company expects to receive). ECLs are discounted at the effective interest rate of the financial asset.

③ Credit-impaired financial assets

At each reporting date, the Company assesses whether financial assets carried at amortized cost and debt securities at FVOCI are credit-impaired. A financial asset is 'credit-impaired' when one or more events that have a detrimental impact on the estimated future cash flows of the financial asset have occurred.

Evidence that a financial asset is credit-impaired includes the following observable data:

- significant financial difficulty of the borrower or issuer;
- a breach of contract such as a default or being more than 90 days past due;
- the restructuring of a loan or advance by the Company on terms that the Company would not consider otherwise;
- it is probable that the borrower will enter bankruptcy or other financial reorganization; or
- the disappearance of an active market for a security because of financial difficulties.

④ Presentation of allowance for ECL in the statement of financial position

Loss allowances for financial assets measured at amortized cost are deducted from the gross carrying amount of the assets. For debt securities at FVOCI, the loss allowance is charged to profit or loss and is recognized in OCI.

⑤ Write off

The gross carrying amount of a financial asset is written off when the Company has no reasonable expectations of recovering a financial asset in its entirety or a portion thereof. For individual customers, the Company has a policy of writing off the gross carrying amount when the financial asset is 180 days past due based on historical experience of recoveries of similar assets. The Company individually makes an assessment with respect to the timing and amount of write-off based on whether there is a reasonable expectation of recovery. The Company expects no significant recovery from the amount written off. However, financial assets that are written off could still be subject to enforcement activities in order to comply with the Company's procedures for recovery of amounts due.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

5. Operating Segments

(1) The Company has disclosed operating segments in the financial statements in accordance with K-IFRS No. 1108. The Company has one reportable segment, 'Biosimilar manufacturing', as of December 31, 2018.

(2) Amount of revenue for three customers who take over 10% or more of the Company's total revenue is ₩434,956 million for the year ended December 31, 2018 (₩440,951 million for three customers for the year ended December 31, 2017).

(3) Geographic revenue for the years ended December 31, 2018 and 2017 were as follows:

(In thousands of won)

		2018	2017
Domestic	₩	227,949,588	18,610,857
Europe		290,009,678	440,951,087
Other		17,846,535	5,067,390
	₩	535,805,801	464,629,334

6. Financial Instruments

(1) Financial assets

Categories of financial assets as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

		December 31, 2018			December 31, 2017		
		Financial assets at fair value through profit or loss	Financial assets measured at amortized cost	Total	Financial assets at fair value through profit or loss	Financial assets measured at amortized cost	Total
Current assets							
Cash and cash equivalents	₩	-	17,621,905	17,621,905	-	358,088,091	358,088,091
Short-term financial instruments		-	550,000,000	550,000,000	-	-	-
Trade and other receivables		-	121,733,350	121,733,350	-	25,817,150	25,817,150
Short-term loans to employees		-	600,000	600,000	-	-	-
Deposits provided		-	20,279,609	20,279,609	-	22,605,000	22,605,000
Financial assets at fair value through profit or loss		540,080,517	-	540,080,517	-	-	-
		540,080,517	710,234,864	1,250,315,381	-	406,510,241	406,510,241
Non-current liabilities							
Long-term financial instruments		-	7,000	7,000	-	7,000	7,000
Deposits provided		-	24,230,355	24,230,355	-	20,560,000	20,560,000
		-	24,237,355	24,237,355	-	20,567,000	20,567,000
	₩	540,080,517	734,472,219	1,274,552,736	-	427,077,241	427,077,241

Samsung Biologics Co., Ltd.
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6. Financial Instruments, Continued

(2) Financial liabilities

Categories of financial liabilities as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

		December 31, 2018			December 31, 2017		
		Financial liabilities at fair value through profit or loss	Financial liabilities measured at amortized cost	Total	Financial liabilities through profit or loss	Financial liabilities measured at amortized cost	Total
Current liabilities							
Trade and other payables	₩	-	170,338,294	170,338,294	-	158,214,846	158,214,846
Derivative liabilities(*1)		-	-	-	1,933,562,427	-	1,933,562,427
Bonds		-	50,000,000	50,000,000	-	49,926,578	49,926,578
Short-term borrowings		-	90,000,000	90,000,000	-	40,000,000	40,000,000
Current portion of long-term borrowings		-	246,583,355	246,583,355	-	48,750,000	48,750,000
		-	556,921,649	556,921,649	1,933,562,427	296,891,424	2,230,453,851
Non-current liabilities							
Trade and other payables		-	3,988,729	3,988,729	-	2,650,638	2,650,638
Bonds		-	189,638,752	189,638,752	-	50,000,000	50,000,000
Long-term borrowings		-	296,474,000	296,474,000	-	357,235,570	357,235,570
Derivative liabilities(*2)		190,330	-	190,330	-	-	-
		190,330	490,101,481	490,291,811	-	409,886,208	409,886,208
	₩	190,330	1,047,023,130	1,047,213,460	1,933,562,427	706,777,632	2,640,340,059

(*1) Biogen Therapeutics Inc. exercised the call option for shares of Samsung Bioepis Co., Ltd. on June 29, 2018 and, all processes for the sale and purchase transaction of shares were completed on November 7, 2018. Therefore, the amount of derivative liabilities is ₩ 0 on December 31, 2018.

(*2) Details of the derivative liabilities that entered into currency swap agreements about foreign currency borrowings, period ended December 31, 2018 are as follows:

Classification	Category	Financial institution	Contracted amount	Contract date	Maturity date	Currency		Contracted exchange rate	Interest rate(%)	
						Sell	Buy		Receiving	Paying
Trading	Currency interest swap	KEB Hana Bank	USD 30,000,000	May 31, 2018	May 31, 2023	USD	KRW	1,080.00	Libor(3M) +1.70	3.70

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
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6. Financial Instruments, Continued

(3) The fair values of financial instruments

The fair values of financial instruments, together with the carrying amounts shown in the statement of financial position as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

	December 31, 2018		December 31, 2017	
	Carrying amount	Fair value	Carrying amount	Fair value
Assets measured at fair value				
Financial assets at fair value through profit or loss	₩ 540,080,517	540,080,517	-	-
Assets not measured fair value				
Cash and cash equivalents	17,621,905	17,621,905	358,088,091	358,088,091
Short-term financial instruments	550,000,000	550,000,000	-	-
Long-term financial instruments	7,000	7,000	7,000	7,000
Trade and other receivables	121,733,350	121,733,350	25,817,150	25,817,150
Short-term loans to employee	600,000	600,000	-	-
Deposits provided	44,509,965	44,509,965	43,165,000	43,165,000
	<u>734,472,220</u>	<u>734,472,220</u>	<u>427,077,241</u>	<u>427,077,241</u>
Total	₩ <u>1,274,552,737</u>	<u>1,274,552,737</u>	<u>427,077,241</u>	<u>427,077,241</u>
Liabilities carried at fair value				
Derivative liabilities	₩ 190,330	190,330	1,933,562,427	1,933,562,427
Liabilities carried at amortized cost				
Trade and other payables	174,327,024	174,327,024	160,865,484	160,865,484
Bonds	239,638,752	239,638,752	99,926,578	99,926,578
Borrowings	633,057,355	633,057,355	445,985,570	445,985,570
	<u>1,047,023,131</u>	<u>1,047,023,131</u>	<u>706,777,632</u>	<u>706,777,632</u>
Total	₩ <u>1,047,213,461</u>	<u>1,047,213,461</u>	<u>2,640,340,059</u>	<u>2,640,340,059</u>

(4) Fair value hierarchy

(i) The fair value hierarchy of financial instruments carried at fair value as of December 31, 2018 is as follows:

(In thousands of won)

	Level 1	Level 2	Level 3	Total
Financial assets at fair value through profit or loss	₩ -	540,080,517	-	540,080,517
Derivative liabilities	-	190,330	-	190,330

(ii) The fair value hierarchy of financial instruments carried at fair value as of December 31, 2017 is as follows:

(In thousands of won)

	Level 1	Level 2	Level 3	Total
Derivative liabilities	₩ -	-	1,933,562,427	1,933,562,427

Samsung Biologics Co., Ltd.
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6. Financial Instruments, Continued

(5) Valuation techniques and inputs

(i) Valuation techniques and significant inputs used in measuring the fair value of financial instruments classified as level 2 and level 3 as of December 31, 2018 are as follows:

(In thousands of won)

		<u>Fair Value</u>	<u>Valuation techniques</u>	<u>Significant unobservable inputs</u>
Financial assets at fair value through profit or loss	₩	540,080,517	Market approach	-
Derivative liabilities		190,330	Discount cash flow model	Currency forward rate, Discount rate, and etc.

(ii) Valuation techniques and significant inputs used in measuring the fair value of financial instruments classified as level 3 as of December 31, 2017 are as follows:

(In thousands of won)

		<u>Fair Value</u>	<u>Valuation techniques</u>	<u>Significant unobservable inputs</u>
Derivative liabilities	₩	1,933,562,427	Binomial model	Fair value of underlying asset (Samsung Bioepis Co., Ltd.)

(6) Finance income and expenses by categories

Details of finance income and expenses for the years ended December 31, 2018 and 2017 are as follows:

(In thousands of won)

	<u>Interest income</u>		<u>Gain (loss) on valuation of derivatives</u>		<u>Other income</u>		<u>Net income</u>	
	<u>(expense)</u>				<u>(expense) (*)</u>		<u>(loss)</u>	
	<u>2018</u>	<u>2017</u>	<u>2018</u>	<u>2017</u>	<u>2018</u>	<u>2017</u>	<u>2018</u>	<u>2017</u>
Financial assets at fair value through profit or loss	-	-	80,517	-	301,927	-	382,444	-
Financial assets measured at amortized cost	9,651,186	9,949,872	-	-	1,500,490	(9,359,709)	11,151,676	590,163
Financial liabilities at fair value through profit or loss	-	-	(190,330)	(59,091,484)	-	-	(190,330)	(59,091,484)
Financial liabilities measured at amortized cost	(15,838,589)	(5,683,983)	-	-	(521,627)	11,059,747	(16,360,216)	5,375,764
	₩ <u>(6,187,403)</u>	<u>4,265,889</u>	<u>(109,813)</u>	<u>(59,091,484)</u>	<u>1,280,790</u>	<u>1,700,038</u>	<u>(5,016,426)</u>	<u>(53,125,557)</u>

(*) Other income and expenses include gain (loss) on foreign currency translations and transactions and gain on disposal of financial assets at fair value through profit or loss.

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7. Cash and Cash Equivalents

Cash and cash equivalents as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

		<u>December 31, 2018</u>	<u>December 31, 2017</u>
Demand deposits	₩	17,621,905	358,088,091

8. Restricted Financial Instruments

Restriction of financial instruments as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

		<u>December 31, 2018</u>	<u>December 31, 2017</u>	<u>Note</u>
Long-term financial instruments	₩	7,000	7,000	Deposits on checking accounts

9. Trade and Other Receivables

(1) Details of trade and other receivables as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

		<u>December 31, 2018</u>			<u>December 31, 2017</u>		
		<u>Related parties</u>	<u>Others</u>	<u>Total</u>	<u>Related parties</u>	<u>Others</u>	<u>Total</u>
Trade receivables	₩	65,408,494	47,147,376	112,555,870	2,503,684	16,015,102	18,518,786
Other receivables		121,565	5,159,933	5,281,498	1,165,506	5,296,165	6,461,671
Accrued income		799,878	3,096,104	3,895,982	579,605	257,088	836,693
	₩	<u>66,329,937</u>	<u>55,403,413</u>	<u>121,733,350</u>	<u>4,248,795</u>	<u>21,568,355</u>	<u>25,817,150</u>

(2) The aging analyses of trade and other receivables that past due but not impaired as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

		<u>December 31, 2018</u>	<u>December 31, 2017</u>
Less than 1 month (*)	₩	85,180,228	24,762,078
More than 1 month, less than 6 months (*)		36,173,535	1,054,998
More than 6 months, less than 12 months (*)		172,423	72
More than 12 months (*)		207,164	2
	₩	<u>121,733,350</u>	<u>25,817,150</u>

(*) It means that the period has passed from the day of receivables issued.

The Company calculated an estimate of expected credit losses based on actual credit loss experience over the past two years and determined that it is not necessary to recognize allowance for trade and other receivables.

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10. Contract Assets and Liabilities

(1) Changes in contract assets and liabilities arising from contracts with customers for the year ended of December 31, 2018 are as follows:

	2018				
(In thousands of won)	Beginning balance	Application of K-IFRS No.1115	Increase	Decrease	Ending balance
Contract assets - costs to fulfil a contract	₩ -	42,152,599	86,561,766	(32,970,495)	95,743,870
Contract assets - unbilled	-	-	834,768	-	834,768
Contract liabilities	-	57,440,382	72,449,486	(28,785,078)	101,104,790

Costs incurred in fulfilling a contract are consisted of costs incurred for tech transfer to manufacture products that customers request and producing trial batches. Such activities do not transfer goods or services to a customer but are required in advance for production of goods ordered. Accordingly, they do not consist as part of a performance obligation but are recognized as contract assets. Unbilled contract assets were recognized due to a difference between the extent of the Company's satisfaction of its performance obligation recognized in progress and the consideration paid by the customer. As the Company satisfies a performance obligation over time, any advance consideration received from a customer are recognized as contract liabilities.

(2) Impairment loss of ₩18,641,061 thousand on contract assets is recognized during the year ended of December 31, 2018.

11. Inventories

(1) Details of inventories as of December 31, 2018 are as follows:

		December 31, 2018		
		Acquisition cost	Provision for loss on valuation	Carrying amounts
Finished goods	₩	84,434,000	(6,442,855)	77,991,145
Work-in-process		77,594,002	(424,700)	77,169,302
Raw materials		83,483,857	-	83,483,857
Supplies		63,716,411	-	63,716,411
	₩	309,228,270	(6,867,555)	302,360,715

(2) Details of inventories as of December 31, 2017 are as follows:

		December 31, 2017		
		Acquisition cost	Provision for loss on valuation	Carrying amounts
Finished goods	₩	104,077,102	(8,793,628)	95,283,474
Work-in-process		22,128,639	(6,770)	22,121,869
Raw materials		45,196,129	-	45,196,129
Supplies		47,385,245	-	47,385,245
	₩	218,787,115	(8,800,398)	209,986,717

(3) During the years ended December 31, 2018 and 2017, provision for loss on valuation of ₩1,932,843 thousand and ₩2,038,561 thousand was reversed respectively. The amounts are included in cost of revenue.

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12. Other Financial Instruments

Details of other financial instruments as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

		December 31, 2018		December 31, 2017	
		Current	Non-current	Current	Non-current
Deposits provided	₩	20,279,609	24,230,355	22,605,000	20,560,000
loans to employee		600,000	-	-	-
Financial assets at fair value through profit or loss		540,080,517	-	-	-
	₩	<u>560,960,126</u>	<u>24,230,355</u>	<u>22,605,000</u>	<u>20,560,000</u>

13. Other Assets

Details of other assets as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

		December 31, 2018		December 31, 2017	
		Current	Non-current	Current	Non-current
Prepaid income taxes	₩	-	-	17,076	-
Advance payments		-	-	140,251	-
Prepaid expenses		5,508,443	1,902,973	4,340,850	963,402
	₩	<u>5,508,443</u>	<u>1,902,973</u>	<u>4,498,177</u>	<u>963,402</u>

14. Investments in Associates and Joint Ventures

(1) Investments in associates and joint ventures as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

Investee	Major Business location	Primary business	Fiscal date	December 31, 2018		December 31, 2017	
				Book value	Percentage of ownership (*1)	Book value	Percentage of ownership
Archigen Biotech Limited	The United Kingdom	Bio-medicine business	December 31, 2018	₩ -	50.00%	10,060,921	50.00%
Samsung Bioepis Co., Ltd.(*1)(*2)	The Republic of Korea	Biosimilar research and commercialization	December 31, 2018	<u>2,561,802,521</u>	50.00%	<u>4,998,173,084</u>	94.61%
				<u>₩ 2,561,802,521</u>		<u>5,008,234,005</u>	

(*1) Biogen Therapeutics Inc. exercised the right to purchase shares (call option) of Samsung Bioepis Co., Ltd., an investee of Samsung Biologics Co., Ltd., up to an ownership of 50%-1 share on June 29, 2018. Accordingly, the Company's stake before the exercise was 94.61% and the stake after the exercise is 50%+1.

(*2) Samsung Bioepis Co., Ltd. was set up with Biogen Therapeutics Inc. in 2012. Samsung Bioepis Co., Ltd. had been classified as a subsidiary until 2014. During 2015, as potential voting rights held by Biogen Therapeutics Inc. were changed to substantive rights to Samsung Bioepis Co., Ltd., the Company lost control over Samsung Bioepis Co., Ltd. According to a joint venture agreement between the Company and Biogen Therapeutics Inc., Samsung Bioepis Co., Ltd. entered into a contract for intellectual property and others with Biogen Therapeutics Inc. and other companies. The Company considered such contractual relationship to determine a control over the investee.

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14. Investments in Associates and Joint Ventures, Continued

As Biogen Therapeutics Inc. exercised the call option for shares of Samsung Bioepis Co., Ltd. on June 29, 2018, and all legal proceedings on trading in share have been completed on the closing date of stock trading (November 7, 2018).

(2) Changes in investments in associates and joint ventures for the year ended December 31, 2018 and 2017 are as follows:

(In thousands of won)

		2018						
Investee		Beginning balance	Acquisition	Equity loss	Equity change	Application of K-IFRS No.1115	Disposals	Ending balance
Archigen Biotech Limited(*)	₩	10,060,921	16,405,000	(27,852,433)	1,386,512	-	-	-
Samsung Bioepis Co., Ltd.		4,998,173,084	-	(101,484,900)	(746,384)	(34,300,467)	(2,299,838,812)	2,561,802,521
	₩	<u>5,008,234,005</u>	<u>16,405,000</u>	<u>(129,337,333)</u>	<u>640,128</u>	<u>(34,300,467)</u>	<u>(2,299,838,812)</u>	<u>2,561,802,521</u>

(*) The Company discontinued the application of the equity method for investment in Archigen Biotech Limited due to the carrying amounts of company's share being reduced to zero during 2018. The unrecognized share of losses due to the suspension of equity method accounting during 2018 amounts to ₩8,328,897 thousand.

(In thousands of won)

		2017				
Investee		Beginning balance	Acquisition	Equity loss	Equity change	Ending balance
Archigen Biotech Limited	₩	48,832,293	-	(31,664,555)	(7,106,817)	10,060,921
Samsung Bioepis Co., Ltd.		4,895,523,997	200,014,156	(97,937,783)	572,714	4,998,173,084
	₩	<u>4,944,356,290</u>	<u>200,014,156</u>	<u>(129,602,338)</u>	<u>(6,534,103)</u>	<u>5,008,234,005</u>

(3) Financial information of equity-accounted investees as of and for the years ended December 31, 2018 and 2017 is summarized as follows:

(In thousands of won)

		December 31, 2018						
Investee		Current assets	Non-current assets	Current liabilities	Non-current liabilities	Revenue	Net loss	Total comprehensive loss
Archigen Biotech Limited	₩	11,705,529	176,477	28,512,961	26,840	-	(95,440,595)	(95,440,595)
Samsung Bioepis Co., Ltd.(*)		1,332,991,753	564,198,972	1,229,279,243	265,777,152	368,733,526	(54,956,629)	(56,585,151)

(*) The amounts are based on consolidated financial information of Samsung Bioepis Co., Ltd. before adjustment of fair value of net assets.

(In thousands of won)

		December 31, 2017						
Investee		Current assets	Non-current assets	Current liabilities	Non-current liabilities	Revenue	Net loss	Total comprehensive loss
Archigen Biotech Limited	₩	24,153,045	75,754,938	19,664,719	1,884,975	-	(63,190,281)	(63,190,281)
Samsung Bioepis Co., Ltd.(*)		547,933,926	637,765,738	453,562,018	237,239,161	314,756,789	(88,346,279)	(87,644,938)

(*) The amounts are based on consolidated financial information of Samsung Bioepis Co., Ltd. before adjustment of fair value of net assets.

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14. Investments in Associates and Joint Ventures, Continued

(4) Reconciliation of the investees' financial information to the carrying amount of the Company's interest in associates and joint ventures as of December 31, 2018 and 2017 is as follows:

(In thousands of won)

December 31, 2018							
Investee	Net assets	Percentage of ownership	Company's share of net assets	Investment difference(*)	Elimination of unrealized profit	Unrecognized equity loss	Carrying amount
Archigen Biotech Limited	₩ (16,657,794)	50.00%	(8,328,897)	-	-	8,328,897	-
Samsung Bioepis Co., Ltd.	402,134,330	50.00%	201,170,693	2,401,335,219	(40,703,391)	-	2,561,802,521

(*) Investment difference includes adjustment of net assets value at point of loss of control and a tentative amount related to irregular capital increase.

(In thousands of won)

	December 31, 2017					
Investee	Net assets	Percentage of ownership	Company's share of net assets	Investment difference(*)	Elimination of unrealized profit	Carrying amount
Archigen Biotech Limited	₩ 78,358,290	50.00%	39,179,145	-	(29,118,224)	10,060,921
Samsung Bioepis Co., Ltd.	495,357,294	94.61 %	468,635,208	4,529,537,876	-	4,998,173,084

(*) Investment difference includes adjustment of net assets value at point of loss of control and a tentative amount related to irregular capital increase.

15. Property, Plant and Equipment

(1) Details of property, plant and equipment as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

December 31, 2018			
	Acquisition cost	Accumulated depreciation	Book value
Buildings	₩ 767,851,048	(42,227,547)	725,623,501
Structures	7,731,647	(660,340)	7,071,307
Machinery	1,100,464,489	(195,747,495)	904,716,994
Vehicles	959,580	(582,548)	377,032
Other tangible assets	39,826,906	(17,010,526)	22,816,380
Construction-in-progress	57,115,774	-	57,115,774
₩	1,973,949,444	(256,228,456)	1,717,720,988

(In thousands of won)

December 31, 2017			
	Acquisition cost	Accumulated depreciation	Book value
Buildings	₩ 615,351,047	(25,952,623)	589,398,424
Structures	7,731,647	(467,049)	7,264,598
Machinery	729,333,461	(132,970,776)	596,362,685
Vehicles	915,080	(428,264)	486,816
Other tangible assets	22,265,352	(11,289,819)	10,975,533
Construction-in-progress	308,844,562	-	308,844,562
₩	1,684,441,149	(171,108,531)	1,513,332,618

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15. Property, Plant and Equipment, Continued

(2) The changes in property, plant and equipment for the year ended December 31, 2018 and 2017 are as follows:

(In thousands of won)

		2018					
		Beginning balance	Acquisition	Disposals	Depreciation	Others	Ending balance
Buildings	₩	589,398,424	7,373,168	-	(15,856,564)	144,708,473	725,623,501
Structures		7,264,598	-	-	(193,291)	-	7,071,307
Machinery		596,362,685	3,224,504	(608,528)	(63,364,286)	369,102,619	904,716,994
Vehicles		486,816	44,500	-	(154,284)	-	377,032
Other tangible assets		10,975,533	4,407,724	(6,368)	(5,953,937)	13,393,428	22,816,380
Construction- in-progress		308,844,562	275,484,587	-	-	(527,213,375)	57,115,774
	₩	<u>1,513,332,618</u>	<u>290,534,483</u>	<u>(614,896)</u>	<u>(85,522,362)</u>	<u>(8,855)</u>	<u>1,717,720,988</u>

(In thousands of won)

		2017					
		Beginning balance	Acquisition	Disposals	Depreciation	Others	Ending balance
Buildings	₩	277,978,463	3,516,171	-	(9,069,308)	316,973,098	589,398,424
Structures		7,200,057	-	-	(188,029)	252,570	7,264,598
Machinery		561,450,254	446,636	(2,727)	(50,527,847)	84,996,369	596,362,685
Vehicles		105,562	463,385	-	(82,131)	-	486,816
Other tangible assets		11,519,970	2,301,426	(2,219)	(4,321,165)	1,477,521	10,975,533
Construction- in-progress		232,705,460	479,820,689	-	-	(403,681,587)	308,844,562
	₩	<u>1,090,959,766</u>	<u>486,548,307</u>	<u>(4,946)</u>	<u>(64,188,480)</u>	<u>17,971</u>	<u>1,513,332,618</u>

(3) The Company recorded construction-in-progress in the amount of ₩152,923,358 thousand and ₩377,714,347 thousand in connection with the construction of buildings and manufacturing facilities for the years ended December 31, 2018 and 2017 respectively. In addition, ₩8,911,242 thousand and ₩10,182,402 thousand of construction-in-progress has been added from borrowing costs capitalization for the years ended December 31, 2018 and 2017 respectively. Interest rate used to calculate the borrowing costs capitalization is 3.45% and 3.17% for the years ended December 31, 2018 and 2017, respectively.

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Notes to the Financial Statements
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16. Intangible Assets

(1) Details of intangible assets as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

December 31, 2018				
	Acquisition cost	Accumulated amortization and impairment loss	Government grants	Book value
Software	₩ 35,350,378	(20,503,970)	-	14,846,408
Other intangible assets	1,059,446	(816,790)	-	242,656
Memberships	2,573,188	(10,756)	-	2,562,432
Land use rights	69,525,088	(10,660,513)	(58,864,575)	-
Construction-in-progress	2,330,138	-	-	2,330,138
	₩ <u>110,838,238</u>	<u>(31,992,029)</u>	<u>(58,864,575)</u>	<u>19,981,634</u>

(In thousands of won)

December 31, 2017				
	Acquisition cost	Accumulated amortization and impairment loss	Government grants	Book value
Software	₩ 27,245,173	(14,742,318)	-	12,502,855
Other intangible assets	1,059,446	(650,249)	-	409,197
Memberships	2,562,188	-	-	2,562,188
Land use rights	69,525,088	(9,270,011)	(60,255,077)	-
Construction-in-progress	3,524,763	-	-	3,524,763
	₩ <u>103,916,658</u>	<u>(24,662,578)</u>	<u>(60,255,077)</u>	<u>18,999,003</u>

16. Intangible Assets, continued

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(2) Changes in intangible assets for the years ended December 31, 2018 and 2017 are as follows:

(In thousands of won)

		2018						
		Beginning balance	Acquisition	Impairment loss	Amortization	Amortization of government grants	Others	Ending balance
Software	₩	12,502,855	2,087,345	-	(5,761,652)	-	6,017,860	14,846,408
Other intangible assets		409,197	-	-	(166,541)	-	-	242,656
Memberships		2,562,188	11,000	(10,756)	-	-	-	2,562,432
Land use rights		-	-	-	(1,390,502)	1,390,502	-	-
Construction-in-progress		3,524,763	4,814,380	-	-	-	(6,009,005)	2,330,138
	₩	<u>18,999,003</u>	<u>6,912,725</u>	<u>(10,756)</u>	<u>(7,318,695)</u>	<u>1,390,502</u>	<u>8,855</u>	<u>19,981,634</u>

(In thousands of won)

		2017						
		Beginning balance	Acquisition	Disposal	Amortization	Amortization of government grants	Others	Ending balance
Software	₩	10,609,884	3,658,536	(39,600)	(5,567,397)	-	3,841,432	12,502,855
Other intangible assets		512,067	67,480	-	(170,350)	-	-	409,197
Memberships		2,062,188	500,000	-	-	-	-	2,562,188
Land use rights		-	-	-	(1,390,502)	1,390,502	-	-
Construction-in-progress		860,515	6,523,651	-	-	-	(3,859,403)	3,524,763
	₩	<u>14,044,654</u>	<u>10,749,667</u>	<u>(39,600)</u>	<u>(7,128,249)</u>	<u>1,390,502</u>	<u>(17,971)</u>	<u>18,999,003</u>

(3) The Company conducted an impairment test for intangible assets with indefinite useful lives and recognized an impairment loss of ₩ 10,756 thousand for memberships during the current term.

17. Debentures issued and Borrowings

(1) Debentures issued and borrowings as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

		December 31, 2018		December 31, 2017	
		Current	Non-current	Current	Non-current
Debentures issued	₩	50,000,000	189,638,752	49,926,578	50,000,000
Short-term borrowings		90,000,000	-	40,000,000	-
Long-term borrowings		246,583,355	296,474,000	48,750,000	357,235,570
	₩	<u>386,583,355</u>	<u>486,112,752</u>	<u>138,676,578</u>	<u>407,235,570</u>

Samsung Biologics Co., Ltd.
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For the years ended December 31, 2018 and 2017

17. Debentures issued and Borrowings, Continued

(2) Details of debentures issued as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

	<u>Issuance date</u>	<u>Maturity date</u>	<u>Interest rate (%)</u>	<u>December 31, 2018</u>	<u>December 31, 2017</u>
Short-term debentures issued					
The 5th placement debenture	Dec 21, 2017	Dec 21, 2018	2.43	₩ -	50,000,000
(Discount on debentures)				-	(73,422)
				<u>-</u>	<u>49,926,578</u>
Long-term debentures issued					
The 1st placement debenture	Oct 24, 2014	Oct 24, 2019	3.48	50,000,000	50,000,000
The 6-1st placement debenture	April 27, 2018	April 27, 2021	3.10	150,000,000	-
The 6-2nd placement debenture	April 27, 2018	April 27, 2023	3.53	40,000,000	-
(Discount on debentures)				(361,248)	-
				<u>239,638,752</u>	<u>50,000,000</u>
(Current portion of long-term borrowings)				(50,000,000)	-
				<u>₩ 189,638,752</u>	<u>99,926,578</u>

(3) Details of short-term borrowings as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

<u>Financial institution</u>	<u>Description</u>	<u>Maturity date</u>	<u>Interest rate (%)</u>	<u>December 31, 2018</u>	<u>December 31, 2017</u>
Korea Development Bank	Loans for equipment	Sep 28, 2019	CD(91D)+0.90	₩ 40,000,000	40,000,000
Mizuho Bank, Ltd	Loans for equipment	Dec 20, 2019	CD(91D)+0.60	50,000,000	-
				<u>₩ 90,000,000</u>	<u>40,000,000</u>

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17. Debentures issued and Borrowings, Continued

(4) Details of long-term borrowings as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

Financial institution	Description	Maturity date	Interest rate (%)	December 31, 2018	December 31, 2017
Korea Development Bank	Loans for equipment	Jul 24, 2019 Jun 12, 2020 (*) Dec 15, 2020 (*)	3.24~3.57 2.86~2.90 2.86	50,000,000 37,500,000 40,000,000	50,000,000 50,000,000 40,000,000
Export-Import Bank of Korea	Overseas investment loans	Jun 14, 2019	LIBOR+1.50	15,094,350	14,463,900
Kookmin Bank		Aug 11, 2020 (*) Sep 28, 2020 (*)	CD(91D)+1.54 CD(91D)+1.43	43,750,000 60,000,000	50,000,000 30,000,000
KEB Hana Bank		Nov 14, 2019 (*) Feb 26, 2021 (*)	CD(91D)+1.45 CD(91D)+1.60	50,000,000 800,000	50,000,000 800,000
	Loans for Equipment	May 31, 2023	LIBOR+1.70	33,543,000	-
NongHyup Bank		Nov 25, 2019	MOR(1Y)+1.52	50,000,000	50,000,000
Woori Bank		Sep 28, 2020	CD(91D)+1.43	70,000,000	30,000,000
Shinhan Bank		Sep 28, 2020	CD(91D)+1.44	70,000,000	30,000,000
Industrial and Commercial Bank of China, Ltd.	Loans for equipment in foreign currency	Jun 29, 2019	LIBOR+1.50	11,189,005	10,721,670
Sumitomo Mitsui Bank	Overseas investment loans	Mar 20, 2021	LIBOR+1.10	11,181,000	-
	Current portion of long-term borrowings			(246,583,355)	(48,750,000)
				<u>₩ 296,474,000</u>	<u>357,235,570</u>

(*) The borrowings have the condition for installment payment every three months starting from two years before their maturities.

(5) Changes in liabilities arising from financial activities for the years ended December 31, 2018 and 2017 are as follows:

	Current portion of debentures issued	Debentures issued	Short-term borrowings	Current portion of long-term borrowings	Long-term borrowings	Total
(In thousands of won)						
Balance as of January 1, 2018	₩ -	99,926,578	40,000,000	48,750,000	357,235,570	545,912,148
Fluctuations in financial cash flows:						
Proceeds from borrowings	-	-	50,000,000	-	153,112,000	203,112,000
Proceeds from issue of bonds	-	189,579,060	-	-	-	189,579,060
Repayment of borrowings	-	-	-	(18,750,000)	-	(18,750,000)
Repayment of bonds	-	(50,000,000)	-	-	-	(50,000,000)
	-	139,579,060	50,000,000	(18,750,000)	153,112,000	323,941,060
Liquidity transfer	50,000,000	(50,000,000)	-	215,485,570	(215,485,570)	-
Exchange rate effects on foreign currency	-	-	-	1,097,785	1,612,000	2,709,785
Discount on debentures	-	133,114	-	-	-	133,114
Balance as of December 31, 2018	₩ 50,000,000	189,638,752	90,000,000	246,583,355	296,474,000	872,696,107

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17. Debentures issued and Borrowings, Continued

			Current portion of long-term borrowings	Long-term borrowings	Total
(In thousands of won)	Bonds	Short-term borrowings			
Balance as of January 1, 2017	₩ 219,917,877	-	233,776,206	389,363,193	843,057,276
Fluctuations in financial cash flows:					
Proceeds from borrowings	-	40,000,000	-	90,000,000	130,000,000
Proceeds from issue of bonds	50,000,000	-	-	-	50,000,000
Cost of issue of bonds	(75,520)	-	-	-	(75,520)
Repayment of borrowings	-	-	(236,562,523)	(60,537,587)	(297,100,110)
Repayment of bonds	(170,000,000)	-	-	-	(170,000,000)
	<u>(120,075,520)</u>	<u>40,000,000</u>	<u>(236,562,523)</u>	<u>29,462,413</u>	<u>(287,175,630)</u>
Liquidity transfer	-	-	57,629,885	(57,629,885)	-
Exchange rate effects on foreign currency	-	-	(6,093,568)	(3,960,151)	(10,053,719)
Discount on debentures	84,221	-	-	-	84,221
Balance as of December 31, 2017	₩ <u>99,926,578</u>	<u>40,000,000</u>	<u>48,750,000</u>	<u>357,235,570</u>	<u>545,912,148</u>

(6) The repayment schedule of bonds and borrowings as of December 31, 2018 is as follows:

(In thousands of won)

Period	Debentures issued	Short-term borrowings	Long-term borrowings	Total
Jan 1, 2019 ~ Dec 31, 2019	₩ 50,000,000	90,000,000	246,583,355	386,583,355
Jan 1, 2020 ~ Dec 31, 2020	-	-	251,650,000	251,650,000
Jan 1, 2021 ~ Dec 31, 2021	150,000,000	-	11,281,000	161,281,000
Jan 1, 2022 ~ Dec 31, 2022	-	-	-	-
Jan 1, 2023 ~	40,000,000	-	33,543,000	73,543,000
	₩ <u>240,000,000</u>	<u>90,000,000</u>	<u>543,057,355</u>	<u>873,057,355</u>

18. Trade and Other Payables

Details of trade and other payables as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

		December 31, 2018		December 31, 2017	
		Current	Non-current	Current	Non-current
Trade payables	₩	1,165,581	-	14,479,382	-
Other payables (*)		119,396,206	-	115,065,186	-
Accrued expenses		49,776,507	3,988,729	28,670,278	2,650,638
	₩	170,338,294	3,988,729	158,214,846	2,650,638

(*) Other payables to related parties as of December 31, 2018 and December 31, 2017 are ₩82,709,228 thousand and ₩88,500,736 thousand, respectively

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19. Emission liability

(1) Emissions rights

The Company has been participating in Emission Trading Scheme in accordance with the Act on Allocation and Trading of Greenhouse Gas Emission since 2017. For 2018, the Company has been granted free of charge emission rights of 56,525 tons of greenhouse gas emissions, and there are no emission rights provided as collateral as of December 31, 2018.

(2) Emission liability

(i) Changes in emission liability for the year ended December 31, 2018 are as follows:

(In thousands of won)

		Beginning balance	Increase	Decrease	Ending balance
Emission liability	₩	502,080	1,493,982	(502,080)	1,493,982

(ii) Changes in emission liability for the year ended December 31, 2017 are as follows:

(In thousands of won)

		Beginning balance	Increase	Decrease	Ending balance
Emission liability	₩	-	502,080	-	502,080

The Company estimated that it emitted 116,284 tons of carbon dioxide for the year ended December 31, 2018.

20. Other Current Liabilities

Details of other current liabilities as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

		December 31, 2018	December 31, 2017
Advance receipts	₩	85,239,342	53,896,457
Withholdings		2,456,272	2,589,718
	₩	<u>87,695,614</u>	<u>56,486,175</u>

21. Employee Benefits

(1) Recognized liabilities for employee benefits as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

	Explanation	December 31, 2018	December 31, 2017
Defined benefit liabilities	Present value of defined benefit obligations	₩ 49,342,672	33,888,400
	Fair value of plan assets	(36,076,367)	(29,892,336)
		<u>13,266,305</u>	<u>3,996,064</u>
Trade and other payables	Accrued expenses (*)	25,410,702	22,021,288
	Long-term accrued expenses (*)	3,988,729	2,650,638
		<u>29,399,431</u>	<u>24,671,926</u>
		₩ <u>42,665,736</u>	<u>28,667,990</u>

(*) The amounts include the accrued compensation for annual leaves, performance bonuses, and others payable to employees.

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Notes to the Financial Statements
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21. Employee Benefits, Continued

(2) Changes in defined benefit liabilities for the years ended December 31, 2018 and 2017 are as follows:

(In thousands of won)

	2018		
	Defined benefit obligations	Plan assets	Net defined benefit liabilities
Beginning balance as of January 1	₩ 33,888,400	(29,892,336)	3,996,064
Included in profit or loss (*)			
Current service cost	11,487,263	-	11,487,263
Interest expense (income)	1,130,987	(1,019,102)	111,885
	<u>12,618,250</u>	<u>(1,019,102)</u>	<u>11,599,148</u>
Remeasurement loss (gain) :			
- Actuarial losses (gains) arising from			
Financial assumptions	2,036,754	-	2,036,754
Experience adjustment	2,561,008	-	2,561,008
- Return on plan assets excluding interest income	-	587,075	587,075
	<u>4,597,762</u>	<u>587,075</u>	<u>5,184,837</u>
Other :			
Contributions to plan assets	-	(6,638,266)	(6,638,266)
Retirement benefits paid	(2,617,301)	1,711,248	(906,053)
Transfer from affiliated companies	855,561	(824,986)	30,575
	<u>(1,761,740)</u>	<u>(5,752,004)</u>	<u>(7,513,744)</u>
Ending balance as of December 31	₩ <u>49,342,672</u>	<u>(36,076,367)</u>	<u>13,266,305</u>

(*) During the year ended December 31, 2018, ₩1,528,264 thousand of retirement benefit was transferred to Construction-in-progress.

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Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

21. Employee Benefits, Continued

(In thousands of won)

	2017		
	Defined benefit obligations	Plan assets	Net defined benefit liabilities
Beginning balance as of January 1	₩ 23,333,875	(17,576,610)	5,757,265
Included in profit or loss (*)			
Current service cost	8,545,737	-	8,545,737
Interest expense (income)	711,053	(543,512)	167,541
	<u>9,256,790</u>	<u>(543,512)</u>	<u>8,713,278</u>
Remeasurement loss (gain) :			
- Actuarial losses (gains) arising from			
Financial assumptions	(170,311)	-	(170,311)
Experience adjustment	2,696,084	-	2,696,084
- Return on plan assets excluding interest income	-	362,898	362,898
	<u>2,525,773</u>	<u>362,898</u>	<u>2,888,671</u>
Other :			
Contributions to plan assets	-	(13,085,879)	(13,085,879)
Retirement benefits paid	(1,551,339)	1,184,968	(366,371)
Transfer from affiliated companies	323,301	(234,201)	89,100
	<u>(1,228,038)</u>	<u>(12,135,112)</u>	<u>(13,363,150)</u>
Ending balance as of December 31	₩ <u>33,888,400</u>	<u>(29,892,336)</u>	<u>3,996,064</u>

(*) During the year ended December 31, 2017, ₩1,075,679 thousand of retirement benefit was transferred to Construction-in-progress.

(3) Details of plan assets as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

	December 31, 2018	December 31, 2017
Cash and cash equivalents	₩ 36,076,367	29,892,336

(4) Defined benefit liabilities

(i) Principal actuarial assumptions as of December 31, 2018 and 2017 are as follows:

	December 31, 2018	December 31, 2017
Discount rate	3.10%	3.60%
Future salary increases	6.05%	6.12%

In order to calculate present value of defined benefit obligations, the Company determined discount rate based on return of corporate bonds, which are consistent with currency of defined benefit obligations and expected payment term as of the reporting date.

As of December 31, 2018 and 2017, the weighted-average duration of the defined benefit obligation is 8.10 years and 7.77 years, respectively.

Samsung Biologics Co., Ltd.
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21. Employee Benefits, Continued

(ii) As of December 31, 2018, effects on present value of defined benefit obligations if each of significant actuarial assumptions changes within potential reasonable range are as follows (if changed by 1%):

(In thousands of won)

		Salary increases		Discount rate	
		1% Up	1% Down	1% Up	1% Down
Present value of defined benefit obligations	₩	4,626,528	(3,981,565)	(3,912,810)	4,629,847

22. Share Capital and Reserves

(1) Details of share capital as of December 31, 2018 and 2017 are summarized as follows:

(In thousands of won, except par value and number of shares)

		December 31, 2018	December 31, 2017
Number of shares authorized		500,000,000	500,000,000
Par value (In won)	₩	2,500	2,500
Number of shares issued		66,165,000	66,165,000
Share capital	₩	165,412,500	165,412,500

(2) Changes in the number of shares issued for the years ended December 31, 2018 and 2017 are as follows:

(In number of shares)

	December 31, 2018		December 31, 2017	
	Number of shares issued	Outstanding stock	Number of shares issued	Outstanding stock
Beginning number of shares	66,165,000	66,165,000	66,165,000	66,165,000
Ending number of shares	66,165,000	66,165,000	66,165,000	66,165,000

(3) Details of share premium as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

		December 31, 2018	December 31, 2017
Additional paid-in capital	₩	2,496,583,409	2,496,583,409
Irregular capital increase		(9,270,327)	(9,270,327)
	₩	2,487,313,082	2,487,313,082

(4) Details of accumulated other comprehensive income (loss) as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

		December 31, 2018	December 31, 2017
Defined benefit plan remeasurements	₩	(12,540,227)	(8,610,119)
Equity adjustment arising from investments in associates and joint ventures		5,564,266	4,600,672
	₩	(6,975,961)	(4,009,447)

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22. Share Capital and Reserves, Continued

(5) Appropriations of retained earnings as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

	2018 (Date of Disposition for 2018 : March 22, 2019)	2017 (Date of Disposition for 2017 : March 22, 2018)
Retained earnings before appropriations	₩ 1,509,718,830	1,327,734,757
Unappropriated retained earnings carried over from prior	1,327,734,757	1,424,706,873
Adjustment on initial application of K-IFRS No.1115	(41,974,909)	-
Equity adjustment arising from investments in associates and joint ventures	(150,213)	-
Net income (loss)	224,109,195	(96,972,116)
Appropriations of retained earnings	-	-
Unappropriated retained earnings to be carried forward to subsequent year	₩ 1,509,718,830	1,327,734,757

23. Revenue and Cost of Revenue

Details of revenue and cost of revenue for the years ended December 31, 2018 and 2017 are as follows:

(In thousands of won)

	2018	2017
Revenue:		
Product sales	₩ 528,227,493	375,324,354
Other sales	7,578,308	89,304,980
	₩ 535,805,801	464,629,334
Cost of revenue:		
Cost of product sales	₩ 346,381,759	238,530,759
Cost of other sales	44,264,100	94,298,514
	₩ 390,645,859	332,829,273

Samsung Biologics Co., Ltd.
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24. Selling, General and Administrative Expenses

Details of selling, general and administrative expenses for years ended December 31, 2018 and 2017 are as follows:

(In thousands of won)

	2018	2017
Salaries	₩ 13,867,455	12,791,255
Bonuses	8,985,853	7,282,797
Retirement benefits	1,661,657	1,267,966
Employee welfare	5,655,411	5,849,421
Depreciation	11,791,680	6,486,402
Amortization of intangible assets	2,823,704	3,008,710
Repairs and maintains	132,180	377,432
Supplies expenses	921,345	581,324
Utility expenses	221,958	213,688
Travel expenses	1,501,657	1,350,522
Service fee	20,541,340	11,514,320
Training expenses	5,795,721	5,228,274
Electronic processing expenses	1,890,192	995,820
Taxes and dues	1,564,795	997,366
Others	12,126,249	7,875,362
	₩ <u>89,481,197</u>	<u>65,820,659</u>

25. Nature of Expenses

Details of expenses classified by nature for the years ended December 31, 2018 and 2017 are as follows:

(In thousands of won)

	2018	2017
Changes in inventories	₩ (115,340,056)	(17,154,371)
Changes in contract assets	6,878,357	-
Raw materials used	154,164,249	75,684,772
Employee benefits	169,729,301	148,689,762
Depreciation and amortization	91,450,555	69,926,227
Supplies expenses	36,881,951	33,346,233
Utility expenses	21,208,597	17,716,247
Service fees	35,680,563	22,787,981
Others (*)	79,473,540	47,653,081
	₩ <u>480,127,057</u>	<u>398,649,932</u>

(*) This amount is the sum of cost of sales and selling, general and administrative expenses in the statements of comprehensive income.

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26. Other Income and Expenses

Details of other income and expenses for the years ended December 31, 2018 and 2017 are as follows:

(In thousands of won)

	2018	2017
Other income		
Rental revenues	₩ 497,992	487,683
Fees revenues	166,509	131,759
Gain on disposal of Property, Plant and Equipment	12,821	-
Gain on disposal of Investments in associates	389,213,672	-
Miscellaneous gains	508,131	270,541
	₩ <u>390,399,125</u>	<u>889,983</u>
Other expenses		
Donations	₩ 13,000	19,900
Loss on disposal of property, plant and equipment	614,509	4,946
Loss on disposal of intangible assets	10,756	39,600
Miscellaneous losses	8,110,008	71,797
	₩ <u>8,748,273</u>	<u>136,243</u>

27. Finance Income and Costs

Details of finance income and costs for the years ended December 31, 2018 and 2017 are as follows:

(In thousands of won)

	2018	2017
Finance income		
Interest income	₩ 9,651,186	9,949,872
Gain on valuation of financial assets at fair value through profit or loss	80,517	-
Gain on disposal of financial assets at fair value through profit or loss	301,927	-
Gain on foreign currency translations	112,646	3,512,447
Gain on foreign currency transactions	9,174,513	9,757,456
	₩ <u>19,320,789</u>	<u>23,219,775</u>
Finance costs		
Interest expenses	₩ 15,838,589	5,683,983
Loss on foreign currency translations	3,291,913	331,259
Loss on foreign currency transactions	5,016,385	11,238,606
Loss on valuation of derivatives	190,330	59,091,484
	₩ <u>24,337,217</u>	<u>76,345,332</u>

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28. Income Tax Expense (Benefit)

(1) Income tax expense (benefit) for the years ended December 31, 2018 and 2017 is as follows:

(In thousands of won)

	2018	2017
Current income tax expense:		
Current income tax expense	₩ 40,772,131	1,865,764
Deferred tax expense:		
Temporary differences	29,226,438	(21,354,723)
Income taxes recognized directly in equity	8,868,073	466,322
Income tax expense (benefit)	₩ 78,866,642	(19,022,637)

(2) The relationship between tax expense (benefit) and accounting profit (loss) for the year ended December 31, 2018 and 2017 is as follows:

(In thousands of won)

	2018	2017
Profit (loss) before income tax	₩ 302,975,837	(115,994,753)
Income taxes at statutory tax rate	73,320,153	(28,070,730)
Adjustments:		
Non-deductible expense	2,161,758	151,764
Changes in unrecognized deferred tax assets	9,015,013	10,177,594
Tax credit carryforward	(4,940,694)	-
Others	(689,588)	(1,281,265)
Income tax expense (benefit)	₩ 78,866,642	(19,022,637)
Average effective tax rate (*)	26.03%	-

(*) The effective tax rate for the year ended December 31, 2017 has not been calculated since the profit before income tax expense is negative.

(3) Income taxes recognized directly in equity(Accumulated other comprehensive income (loss)) for the year ended December 31, 2018 and 2017 are as follows:

(In thousands of won)

	2018			2017		
	Before tax	Tax expense	Net of tax	Before tax	Tax expense	Net of tax
Defined benefit plan remeasurements	₩ (5,184,837)	1,254,731	(3,930,106)	(2,888,671)	699,058	(2,189,613)
Equity adjustments arising from investments in associates and joint ventures	963,592	-	963,592	(6,534,103)	(232,736)	(6,766,839)
	₩ (4,221,245)	1,254,731	(2,966,514)	(9,422,774)	466,322	(8,956,452)

(4) Income taxes recognized directly in equity (Retained earnings) for the year ended December 31, 2018 and 2017 are as follows:

(In thousands of won)

	2018			2017		
	Before tax	Tax expense	Net of tax	Before tax	Tax expense	Net of tax
Retained earnings (*)	₩ (49,588,251)	7,613,342	(41,974,909)	-	-	-

(*) This occurred as a result of adjusting the cumulative effect of applying K-IFRS No.1115, "Revenue from contracts with customers," to the equity as of January 1, 2018.

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28. Income Tax Expense (Benefit), Continued

(5) Changes in temporary differences and deferred tax assets (liabilities) as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

		2018			
		January 1, 2018	Adjustments of beginning	Profit or loss	December 31, 2018
Accrued expenses	₩	2,685,899	-	1,516,128	-
Net defined benefit obligations		949,957	-	789,894	1,254,731
Long-term accrued expenses		641,454	-	323,819	-
Loss (gain) on foreign currency translations		(579,202)	-	764,407	-
Provision for loss on valuation of finished goods		2,128,058	-	(568,887)	-
Provision for loss on valuation of work in progress		1,638	-	101,139	-
Depreciation		161,942	-	2,733,853	-
Provision for performance bonus		3,862,496	-	449,920	-
Derivative liabilities		467,922,107	-	(467,922,107)	-
Equity-accounted investees		(1,039,776,584)	3,913,698	454,737,111	-
Accrued income		(15,446)	-	(374,307)	-
Emission liability		121,503	-	240,041	-
Contract Assets/Liabilities		-	3,699,644	(2,604,315)	-
Impairment of intangible assets		-	-	2,603	-
Gain on valuation of derivative assets-SWAP		-	-	46,060	-
Tax loss carryforward		33,270,565	-	(33,270,565)	-
Tax credit carryforward		23,309,722	-	4,940,694	-
Deferred tax assets (liabilities)	₩	(505,315,891)	7,613,342	(38,094,512)	1,254,731

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28. Income Tax Expense (Benefit), Continued

(In thousands of won)

		2017		
		January 1, 2017	Profit or loss	Other comprehensive income(loss) December 31, 2017
Accrued expenses	₩	2,132,868	553,031	-
Net defined benefit obligations		1,154,753	(903,854)	699,058
Long-term accrued expenses		556,692	84,762	-
Loss (gain) on foreign currency translations		2,613,108	(3,192,310)	-
Provision for loss on valuation of finished goods		2,035,998	92,060	-
Provision for loss on valuation of work in progress		587,030	(585,392)	-
Depreciation		52,408	109,534	-
Provision for performance bonus		2,768,614	1,093,882	-
Derivative liabilities		453,621,968	14,300,139	-
Equity-accounted investees		(1,052,926,863)	13,383,015	(232,736)
Accrued income		(645,778)	630,332	-
Emission liability		-	121,503	-
Tax loss carryforward		45,877,938	(12,607,373)	-
Tax credit carryforward		15,500,650	7,809,072	-
Deferred tax assets (liabilities)	₩	(526,670,614)	20,888,441	466,322

(6) As of December 31, 2018 and 2017, the Company did not recognize ₩115,935,862 thousand and ₩104,277,416 thousand of deferred tax assets, respectively, with respect to investments in associates and joint ventures, tax credit carryforward, and so forth as the Company determines that it is not probable that they will be realized in foreseeable future.

29. Earnings (Loss) per Share

(1) Basic earnings (loss) per share for the years ended December 31, 2018 and 2017 are calculated as follows:

(In won and number of shares)

		2018	2017
Net profit (loss) for the year	₩	224,109,194,872	(96,972,115,811)
Weighted-average number of ordinary shares outstanding		66,165,000	66,165,000
Basic earnings (loss) per share	₩	3,387	(1,466)

(2) Weighted-average number of ordinary shares outstanding

(In number of shares)

	2018	2017
Issued ordinary shares at the beginning	66,165,000	66,165,000
Weighted-average number of ordinary shares as of December 31	66,165,000	66,165,000

(3) The Company has no dilutive potential shares; therefore, diluted earnings per share is equal to the basic earnings per share.

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30. Commitments and Contingencies

(1) Biogen Therapeutics Inc. held a call option to purchase shares of Samsung Bioepis Co., Ltd., an investee of the Company, up to the ownership of 50%-1 share under the shareholders' agreement and exercised the call option on June 29, 2018 and, all processes for the sale and purchase transaction of shares were completed on November 7, 2018. Therefore, the amount of derivative liabilities is ₩ 0 on December 31, 2018.

(2) As of December 31, 2018, details of the Company's borrowings commitment with financial institutions are summarized as follows:

(In thousands of won, USD)

Financial institution	Remarks	Currency	Limit	Outstanding balance
Woori Bank	Bank overdraft	KRW	10,000,000	-
	B2B PLUS (*)	KRW	7,200,000	7,200,000
	General loan	KRW	30,000,000	-
	Loans for equipment	KRW	70,000,000	70,000,000
KEB Hana Bank	Bank overdraft	KRW	10,000,000	-
	Loans for equipment	KRW	50,000,000	50,000,000
	Loans for equipment	KRW	50,000,000	800,000
	Comprehensive limit	KRW	40,000,000	-
	Loans for equipment in foreign currency	USD	30,000	30,000
Shinhan Bank	General loan	KRW	19,214,000	-
	Bank overdraft	KRW	10,000,000	-
	Loans for equipment	KRW	70,000,000	70,000,000
Kookmin Bank	Loans for equipment	KRW	43,750,000	43,750,000
	Loans for equipment	KRW	60,000,000	60,000,000
NongHyup Bank	Loans for equipment	KRW	50,000,000	50,000,000
Korea Development Bank	Loans for equipment	KRW	50,000,000	50,000,000
	Loans for equipment	KRW	40,000,000	40,000,000
	Loans for equipment	KRW	37,500,000	37,500,000
	Loans for equipment	KRW	40,000,000	40,000,000
	Loans for equipment	KRW	50,000,000	50,000,000
Mizuho Bank, Ltd	Loans for equipment	KRW	50,000,000	50,000,000
Export-Import Bank of Korea	Overseas investment loans	USD	13,500	13,500
Industrial and Commercial Bank of China, Ltd.	Loans for equipment in foreign currency	USD	50,000	10,007
Sumitomo Mitsui Bank	Overseas investment loans	USD	40,000	10,000

(*) The company commit with Woori Bank about B2B transactions, which can execute by the company's customer. The outstanding balance is accounted on other payables.

(3) As of December 31, 2018, details of guarantees provided by financial institutions are summarized as follows:

(In thousands of won, USD)

Financial institution	Remarks	Currency	Limit	Currency	Outstanding balance
Woori Bank	Payment guarantees	KRW	9,500,000	KRW	2,071,579
Shinhan Bank	Letter of credit	USD	10,000	USD	-
	Foreign currency payment guarantee	USD	10,000	USD	-
Kookmin Bank	Foreign currency payment guarantee	KRW	30,000,000	USD	-

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30. Commitments and Contingencies, Continued

(4) Major commitments with financial institutions

If the Company does not comply with the commitments below, financial institutions may require an early redemption of long-term borrowings. Details of debt covenants related to long-term borrowings with financial institutions as of December 31, 2018 are as follows:

Financial institution	Remarks	Details of covenants
Kookmin Bank	Retention of management rights	The Company shall continue to be an affiliate of the Samsung Group, and a current largest shareholder should retain its management rights.
Sumitomo Mitsui Bank	Shareholder composition ratio	The ownership of the Samsung Group shall be over 50%, and Samsung C&T Corporation shall maintain its ownership in the Company in excess of 30%.
Mizuho Bank, Ltd	Shareholder composition ratio	The ownership of the Samsung Group (Samsung C&T Corporation and Samsung Electronics Co., Ltd.) shall be over 60% of the voting rights stock

(5) In connection with a lease agreement with Incheon Metropolitan city ("Incheon") related to the use of land, the Company is exempt from the full amount of rent because the Company meets the requirements for exemption of rent in accordance with the Common Property Management Ordinance of Incheon. The Company recognized an intangible asset at the fair value of the rent that is exempt from Incheon and government grant as a reduction of assets at the same value. The period of land use without payment and exemption requirement of the full amount of rent are as follows:

(i) Period

A lease term is for a total of 50 years. The initial lease period is for 20 years from the land lease contract date, April 28, 2011, and the Company can renew the contract for another 30 years in successive 10 year increments.

(ii) Exemption requirements

The Company whose foreign investment amounts to more than USD 20 million for 5 years since the land lease contract date is subject to the exemption from Common Property Management Ordinance of Incheon. Afterwards, the Company, including foreign invested enterprises that meet the average daily employee count of above 300, is subject to exemption.

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31. Related Parties

(1) As of December 31, 2018, Samsung C&T Corporation is a controlling company, and the percentage of ownership it has over the Company is 43.44%.

(2) Details of related parties as of December 31, 2018 are as follows:

Relationship	Companies
Parent	Samsung C&T Corporation
Entity with significant influence over the Company	Samsung Electronics Co., Ltd.
Associates and Joint ventures	Samsung Bioepis Co., Ltd., Archigen Biotech Limited
Other	Samsung Welstory Inc., Samsung C&T Deutschland GmbH, Samsung Research America, Inc., Samsung Electronics Sales Co., Ltd.
The Enterprise Group (*)	Samsung Card Co., Ltd., Samsung Engineering Co., Ltd. S-1 Corporation, Samsung SDS Co., Ltd., Samsung Economic Research Institute, Multicampus Corporation, Samsung Fire & Marine Insurance Co., Ltd., Samsung Life Insurance Co., Ltd., Hotel Shilla Co., Ltd., Cheil Worldwide Inc., Samsung securities Co., Ltd

(*) The entities are not included in the scope of related parties in accordance with K-IFRS No. 1024, but are included in the scope of Enterprise Group designated by Fair Trade Commission of the Republic of Korea.

(3) Account balances with related parties as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

Relationship	Companies	December 31, 2018					
		Receivables			Payables		
		Trade receivables	Other receivables	Accrued income	Other payables	Advance receipts	Contract liabilities
Entity with significant influence over the Company	Samsung Electronics Co., Ltd. ₩	-	-	-	1,777,735	-	-
Associates and Joint Ventures	Samsung Bioepis Co., Ltd.	65,408,494	106,572	522,506	-	2,077,939	18,289,073
	Archigen Biotech Limited	-	-	277,371	-	-	-
Others	Samsung Welstory Inc.	-	3,014	-	727,587	-	-
	Samsung C&T Deutschland GmbH	-	-	-	80,785	-	-
	Samsung Research America, Inc.	-	-	-	94,194	-	-
The Enterprise Group	Samsung Card Co., Ltd.	-	11,979	-	253,644	-	-
	Samsung Engineering Co., Ltd.	-	-	-	75,343,455	-	-
	S-1 Corporation	-	-	-	1,370,857	-	-
	Samsung SDS Co., Ltd.	-	-	-	2,621,992	-	-
	Samsung Economic Research Institute	-	-	-	322,761	-	-
	Samsung Fire & Marine Insurance Co., Ltd.	-	-	-	960	-	-
	Others(*)	-	-	-	115,258	-	-
		₩ 65,408,494	121,565	799,877	82,709,228	2,077,939	18,289,073

(*) Transactions with Multicampus Corporation and Hotel Shilla Co., Ltd. And Samsung Life Insurance Co., Ltd were included.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

31. Related Parties, Continued

(In thousands of won)

		December 31, 2017				
Relationship	Companies	Receivables			Payables	
		Trade receivables	Other receivables	Accrued income	Other payables	Advance receipts
Entity with significant influence over the Company	Samsung Electronics Co., Ltd.	₩ -	-	-	22,328	-
Associates and Joint ventures	Samsung Bioepis Co., Ltd. (*1)	2,503,684	647,378	52,134	-	4,463,334
Other	Archigen Biotech Limited	-	514,990	527,471	5,161	-
	Samsung Welstory Inc.	-	3,138	-	682,238	-
	Samsung C&T Deutschland GmbH	-	-	-	45,015	-
	Samsung Research America, Inc.	-	-	-	193,902	-
The Enterprise Group	Samsung Card Co., Ltd.	-	-	-	64,884,955	-
	Samsung Engineering Co., Ltd.	-	-	-	16,759,196	-
	S-1 Corporation	-	-	-	3,192,538	-
	Samsung SDS Co., Ltd.	-	-	-	2,333,707	-
	Samsung Economic Research Institute	-	-	-	298,081	-
	Samsung Fire & Marine Insurance Co., Ltd.	-	-	-	2,177	-
	Others (*2)	-	-	-	81,438	-
		₩ 2,503,684	1,165,506	579,605	88,500,736	4,463,334

(*1) The amount of trade receivables reflects offset against advance receipts of ₩65,027 thousand of Samsung Bioepis Co., Ltd.

(*2) Transactions with Multicampus Corporation, Hotel Shilla Co., Ltd., and Cheil Worldwide Inc. were included.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

31. Related Parties, Continued

(4) Significant transactions which occurred in the normal course of business with related parties for the years ended December 31, 2018 and 2017 are summarized as follows:

(In thousands of won)

Relationship	Companies	2018				
		Revenue		Expense		Asset acquisition
		Revenue	Others	Service fees	Others	
Parent	Samsung C&T Corporation	₩ -	-	363,021	-	3,280,317
Entity with significant influence over the Company	Samsung Electronics Co., Ltd.	-	-	-	1,621,550	-
Associates and Joint Ventures	Samsung Bioepis Co., Ltd.	225,444,041	591,158	-	-	-
	Archigen Biotech Limited	-	17,695	-	9,384	-
Others	Samsung Welstory Inc.	-	29,798	-	7,426,091	-
	Samsung C&T Deutschland GmbH	-	-	-	800,507	-
	Samsung Research America, Inc.	-	-	1,758,213	-	-
	Samsung Electronics Sales Co., Ltd.	-	3,600	-	33,368	-
The Enterprise Group	Samsung Card Co., Ltd.	-	7,902	-	313,846	-
	Samsung Engineering Co., Ltd.	-	-	-	-	121,129,703
	S-1 Corporation	-	-	11,114,727	929,556	810,600
	Samsung SDS Co., Ltd.	-	-	6,489	12,525,616	3,652,816
	Samsung Economic Research Institute	-	-	-	1,320,090	-
	Samsung Fire & Marine Insurance Co., Ltd.(*1)	-	412,335	-	3,909,654	-
	Samsung Life Insurance Co., Ltd.(*1)	-	81,300	-	27,877	-
	Others(*2)	-	-	38,352	763,824	-
		₩ 225,444,041	1,143,788	13,280,802	29,681,363	128,873,436

(*1) Other revenue includes actual return on plan assets recognized as deduction of retirement benefits.

(*2) Transactions with Multicampus Corporation and Hotel Shilla Co., Ltd. were included.

(In thousands of won)

Relationship	Companies	2017				
		Revenue		Expense		Asset acquisition
		Revenue	Others	Service fees	Others	
Parent	Samsung C&T Corporation	₩ -	-	143,632	16,992	952,253
Entity with significant influence over the Company	Samsung Electronics Co., Ltd.	-	-	-	27,640	-
Associates and Joint Ventures	Samsung Bioepis Co., Ltd.	18,423,815	542,150	-	-	-
	Archigen Biotech Limited	-	23,722	-	9,384	-
Others	Samsung Welstory Inc.	-	30,057	-	6,710,571	-
	Samsung C&T Deutschland GmbH	-	-	60,479	248,009	-
	Samsung Research America, Inc.	-	-	2,448,390	-	-
	Samsung Electronics Sales Co., Ltd.	-	3,590	-	13,160	-
The Enterprise Group	Samsung Card Co., Ltd.	-	-	-	207,225	-
	Samsung Engineering Co., Ltd.	-	-	-	-	310,898,297
	S-1 Corporation	-	-	6,356,280	528,938	4,283,760
	Samsung SDS Co., Ltd.	-	-	-	10,815,639	6,240,750
	Samsung Economic Research Institute	-	-	1,075,363	1,273,498	-
	Samsung Fire & Marine Insurance Co., Ltd. (*1)	-	297,954	2,357	3,288,498	-
	Others(*2)	-	4,630	119,046	1,405,954	-
		₩ 18,423,815	902,103	10,205,547	24,545,508	322,375,060

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

31. Related Parties, Continued

(*1) Other revenue includes actual return on plan assets recognized as deduction of retirement benefits.

(*2) Transactions with Multicampus Corporation and Hotel Shilla Co., Ltd. were included.

(5) Key management personnel compensation for each of the following categories for the years ended December 31, 2018 and 2017 are summarized as follows:

(In thousands of won)

		2018	2017
Short-term employee benefits	₩	3,419,030	2,314,139
Long-term employee benefits		1,889,461	1,372,888
Retirement benefits		243,788	248,155
	₩	<u>5,552,279</u>	<u>3,935,182</u>

Key management refers to the registered directors and audit committee. Compensation of the Company's key management personnel includes salaries, non-cash benefits, and contributions to a post-employment defined benefit plan. In addition, the Company has granted long-term incentives to the Company's key management personnel based on its three-year management performance. The estimated amount of liabilities arising from such incentives are recognized as accrued expenses and long-term accrued expenses.

(6) The Company made a lease agreement with Incheon Metropolitan city ("Incheon") related to the use of land and subleased the land to Samsung Bioepis Co., Ltd., its associate, in consent of Incheon.

(7) Details of financial transactions with related parties for the years ended December 31, 2018 and 2017 are as follows:

(In thousands of won)

	Companies		2018	2017	Description
Associates and	Archigen Biotech Limited	₩	16,405,000	-	Equity contribution in cash
Joint Ventures	Samsung Bioepis Co., Ltd.		-	200,000,000	Equity contribution in cash
Others	Employee Loan		600,000	-	Loan

32. Statements of Cash Flows

Significant transactions not involving cash flows of investing and financing activities for the years ended December 31, 2018 and 2017 are summarized as follows:

(In thousands of won)

		2018	2017
Increase in other payables due to acquisition of property, plant and equipment	₩	(66,701,472)	18,761,648
Transfer construction-in-progress to property, plant and equipment		533,223,341	407,540,989

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

33. Financial Risk Management

The Company is exposed to credit risk, liquidity risk, and market risk in connection with its financial instruments. This note presents information about the Company's exposure to each of the above risks, the Company's objectives, policies and processes for measuring and managing risks, and the Company's management of capital. Further quantitative disclosures are included throughout these financial statements.

(1) Financial risk management

The Company's risk management policies are established to identify and analyze the risks faced by the Company, to set appropriate risk limits and controls, and to monitor risks and adherence to limits. Risk management policies and systems are reviewed regularly to reflect changes in market conditions and the Company's activities. The Company, through its training and management standards and procedures, aims to maintain a disciplined and constructive control environment in which all employees understand their roles and obligations. The Board of Directors oversees how management monitors compliance with the Company's risk management policies and procedures and reviews the adequacy of the risk management framework in relation to the risks faced by the Company.

(2) Credit risk

Credit risk is the risk of financial loss to the Company if a customer or counterparty to a financial instrument fails to meet its contractual obligations and arises principally from the Company's receivables from customers.

The carrying amount of financial assets represents the maximum credit exposure. The maximum exposure to credit risk as of December 31, 2018 and 2017 are as follows:

(In thousands of won)

	December 31, 2018	December 31, 2017
Cash and cash equivalents	₩ 17,621,905	358,088,091
Short-term financial instruments	550,000,000	-
Short-term loans to employee	600,000	-
Financial assets at fair value through profit or loss	540,080,517	-
Long-term financial instruments	7,000	7,000
Trade and other receivables	121,733,350	25,817,150
Deposits provided	44,509,965	43,165,000
	₩ 1,274,552,737	427,077,241

(3) Liquidity risk

Liquidity risk is the risk that the Company will encounter difficulty in meeting the obligations associated with its financial liabilities that are settled by delivering cash or other financial assets. The Company's approach to managing liquidity is to ensure, as far as possible, that it will always have sufficient liquidity to meet its liabilities when they are due, under both normal and stressed conditions, without incurring unacceptable losses or risking damage to the Company's reputation.

The Company monitors its cash flows through long-term and short-term management strategies and ensures it has sufficient cash on demand to meet expected operational expenditures. This excludes the potential impact of extreme circumstances that cannot reasonably be predicted.

The Company establishes short-term and long-term cash management plans to manage liquidity risk. The Company matches maturity structures of financial assets and liabilities through analyzing and reviewing cash flow budget and actual cash flow. Management believes that the Company is able to redeem its financial liabilities through operating cash flows and cash inflows of financial assets.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

33. Financial Risk Management, Continued

The following is the contractual maturities of financial liabilities as of December 31, 2018 and 2017 including estimated interest payments and excluding the impact of netting agreements. The Company does not expect that the cash flows included in the maturity analysis could occur significantly earlier or at different amounts.

(In thousands of won)

		December 31, 2018					
		Carrying amount	Contractual cash flows	Less than 1 year	1-2 year	2-3 year	More than 3 years
Trade and other payables	₩	174,327,024	174,327,024	170,338,294	3,988,730	-	-
Bonds		239,638,752	258,311,904	57,471,415	6,073,394	152,901,425	41,865,670
Short-term borrowings		90,000,000	92,050,285	92,050,285	-	-	-
Long-term borrowings		543,247,685	567,874,119	262,062,238	259,015,013	12,654,638	34,142,230
	₩	1,047,213,461	1,092,563,332	581,922,232	269,077,137	165,556,063	76,007,900

(In thousands of won)

in thousands of Won)

		December 31, 2017					
		Carrying amount	Contractual cash flows	Less than 1 year	1-2 year	2-3 year	More than 3 years
Trade and other payables	₩	160,865,484	160,865,484	158,214,846	2,650,638	-	-
Bonds		99,926,578	104,407,787	52,993,172	51,414,615	-	-
Short-term borrowings		40,000,000	40,760,285	40,760,285	-	-	-
Long-term borrowings		405,985,570	428,612,072	60,718,045	254,099,571	113,693,973	100,483
	₩	706,777,632	734,645,628	312,686,348	308,164,824	113,693,973	100,483

(4) Currency risk

The Company is exposed to currency risk on sales, purchases, and borrowings. The currencies in which these transactions primarily are denominated are USD and so forth. The Company establishes a plan to manage currency risk through currency forwards and so forth, if considered necessary.

(i) Exposure to currency risk

The Company's exposure to foreign currency risk as of December 31, 2018 and 2017 is as follows:

(In thousands of won)

		December 31, 2018				
		USD	EUR	CHF	JPY	GBP
Assets:						
Trade and other receivables	₩	110,741,934	55,228	-	-	8,467
Contract assets		711,399	-	-	-	-
Liabilities:						
Borrowings		(37,464,355)	-	-	-	-
Trade and other payables		(1,377,411)	(3,671,545)	(108,852)	(20,264)	(62,078)
	₩	<u>72,611,567</u>	<u>(3,616,317)</u>	<u>(108,852)</u>	<u>(20,264)</u>	<u>(53,611)</u>

(In thousands of won)

		December 31, 2017			
		USD	EUR	CHF	JPY
Assets:					
Cash equivalents	₩	21,256,612	-	-	-
Trade and other receivables		20,480,232	14,072	-	-
Liabilities:					
Borrowings		(25,185,570)	-	-	-
Trade and other payables		(10,869,294)	(342,467)	(36,421)	(3,796)
	₩	<u>5,681,980</u>	<u>(328,395)</u>	<u>(36,421)</u>	<u>(3,796)</u>

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

33. Financial Risk Management, Continued

(ii) The following significant exchange rates applied for the years ended December 31, 2018 and 2017 are as follow:

		Average rate		Reporting date spot rate	
		2018	2017	December 31, 2018	December 31, 2017
USD	₩	1,100.30	1,130.84	1,118.10	1,071.40
EUR		1,298.63	1,276.39	1,279.16	1,279.25
CHF		1,124.29	1,148.72	1,136.22	1,094.72
JPY(100 ¥)		996.27	1,008.52	1,013.18	949.11
GBP		1,468.22	1,455.43	1,420.32	1,439.53

(iii) Sensitivity analysis

A strengthening (weakening) of won, a change of 10% in foreign currency exchange rate as indicated below, against EUR, CHF, and USD as of December 31, 2018 and 2017 would have increased (decreased) profit or loss (before tax) by the amounts shown below. This analysis is based on foreign currency exchange rate variances that the Company considered to be reasonably possible at the reporting date.

		December 31, 2018		December 31, 2017	
		10% increase	10% decrease	10% increase	10% decrease
USD	₩	7,261,157	(7,261,157)	568,198	(568,198)
EUR		(361,632)	361,632	(32,840)	32,840
CHF		(10,885)	10,885	(3,642)	3,642
JPY		(2,026)	2,026	(380)	380
GBP		(5,361)	5,361	-	-
	₩	6,881,253	(6,881,253)	531,336	(531,336)

(5) Interest rate risk

The Company is exposed to interest rate risk because they raised funds with interests paid at variable rate. The effect of changes in interest rates by 0.5% applied to the profit and loss (before tax) as of December 31, 2018 and 2017 are as follows:

		December 31, 2018		December 31, 2017	
		0.5% increase	0.5% decrease	0.5% increase	0.5% decrease
Interest expense	₩	(1,973,725)	1,973,725	(1,047,325)	1,047,325

(6) Capital management

The Company's objectives when managing capital are to safeguard the Company's ability to continue as a going concern in order to provide returns for shareholders and benefits for other stakeholders and to maintain an optimal capital structure to reduce the cost of capital.

The debt-to-equity ratio and net borrowing ratio as of December 31, 2018 and 2017 are as follows:

		December 31, 2018	December 31, 2017
Total liabilities (A)	₩	1,824,940,197	3,206,640,270
Total equity (B)		4,155,468,451	3,976,450,892
Cash and cash equivalents (C)		17,621,905	358,088,091
Total bonds and borrowings (D)		872,696,106	545,912,148
Debt-to-equity ratio (A/B)		43.92%	80.64%
Adjusted net borrowing ratio ((D-C)/B)		20.58%	4.72%

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

34. Exercised a call option

Biogen Therapeutics Inc. held a call option to purchase shares in Samsung Bioepis Co., Ltd., an investee of the Company, up to the ownership of 50%-1 share under the shareholders' agreement and exercised the option on June 29, 2018.

And November 7, 2018, all processes for the sale and purchase transaction of shares were completed.

Details of Investments in associates and joint ventures for sale as of annual period 2018 are as follows:

(In thousands of won)

	Amount
Sale closing date	November 7, 2018
Final price for sale	₩ 759,460,612
Related security transaction tax	(3,797,303)
Asset for sale and related liabilities :	
Investments in associates and joint ventures	(2,299,838,812)
Derivatives liabilities	1,933,562,427
Accumulated other comprehensive loss related to asset for sale :	
Equity adjustments arising from investments in associates and joint ventures	(173,252)
Gain on disposal of investments in associates and joint ventures	389,213,672

35. Related to the Securities & Futures Commission's ruling and administrative litigation

On July 12, 2018, the Securities & Futures Commission("SFC") announced its ruling according to the Act on External Audit of Stock Companies against the Company for failing to disclose information regarding the joint venture agreement between Biogen Therapeutics Inc in the public disclosure. Accordingly, the following measures were taken against the Company by the SFC: i) compulsory designation of external auditors for three years; ii) recommendation of dismissal of the Company's executives in charge; and iii) prosecution against the Company and its representative director(CEO).

On November 14, 2018 the SFC announced its ruling according to the Act on External Audit of Stock Companies against the Company for violating accounting standards by inappropriately applying consolidated method regarding investments in Samsung Bioepis. Accordingly, the following measures were taken against the Company by the SFC : i) compulsory designation of external auditors for three years; ii) recommendation of dismissal of the Company's representative director(CEO) and executives in charge; iii) prosecution against the Company and its representative director(CEO) ; iv) retrospective restatement of financial statements; and v) imposition of administrative fine worth ₩ 8 billion. On November 21, 2018 the SFC notified the Company of a fine of ₩ 8 billion.

To justify the accounting procedure of the Company, the Company filed an administrative lawsuit demanding cancellation of the SFC and the Financial Services Commission's action on October 8, 2018 and November 27, 2018 and a lawsuit requesting suspension of the action on November 27, 2018 and December 18, 2018 to the Seoul Administrative Court and the trials are in progress.

The Seoul Administrative Court has suspended the both measures taken by the Securities & Futures Commission ("SFC") against the Company until 30 days after the court makes its final ruling in a lawsuit filed by the Company.

The court has accepted the Company's request on January 22, 2019 which was submitted by the Company on November 27, 2018 to suspend the SFC's following measures against the Company: i) compulsory designation of external auditors for three years; ii) recommendation of dismissal of the Company's representative director(CEO) and executives in charge; and iii) retrospective restatement of financial statements.

35. Related to the Securities & Futures Commission's ruling and administrative litigation, Continued

The court also has accepted the Company's request on February 19, 2019 which was submitted by the Company on December 18, 2018 to suspend the SFC's following measures against the Company: i) compulsory designation of external auditors for three years; and ii) recommendation of dismissal of the Company's executives in charge.

Although the future outcome of the administrative litigation cannot be estimated, if the company loses the administrative litigation, the effect of the restatement of the financial statements is as follows.

If the company restates its financial statements according to the SFC, Investments in associates and joint ventures as of December 31, 2018 are ₩ 174 billion. It is going to decrease by ₩ 2.562 trillion after restatement from ₩ 2.387 trillion before restatement. This is mainly due to the elimination of fair value valuation for Samsung Bioepis in 2015.

And after the restatement, gain on disposal of investments in associates and joint ventures is ₩ 2.508 trillion, which will increase ₩ 2.119 trillion after the restatement from ₩ 389 billion. This is a result of the SFC's points to eliminate the ₩ 4.543 trillion of profit on the fair value of the joints venture in Samsung Bioepis in 2015, but it only reflects ₩ 1.820 trillion in derivative liabilities.

This derivative liabilities as of December 31, 2017 was ₩ 1.933 trillion before the restatement, but it was removed by the close of the transaction on November 7, 2018 following the call option exercise by Biogen Therapeutics. Inc. on June 29, 2018. The amount is zero ('0') regardless of whether it is restated or not.

After the restatement which above effects are reflected, the amount of Income before income tax is ₩ 2.419 trillion which ₩ 2.116 trillion is increased from ₩ 303 billion before the restatement. Net income after the restatement is ₩ 1.827 trillion which ₩ 1.603 trillion is increased from ₩ 224 billion before the restatement.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

35. Related to the Securities & Futures Commission's ruling and administrative litigation, Continued

(1) Impact on Financial State

If the company loses the administrative litigation, it shall rewrite the statements of financial position as indicated by the Securities and Futures Commission and its impact shall be as follows.

(i) In 2018 and 2017

(In thousands of won)

	2018			2017		
	Before rewriting	Adjustment	After rewriting	Before rewriting	Adjustment	After rewriting
Assets						
Current assets	1,577,683,716	-	1,577,683,716	620,995,135	-	620,995,135
Non-current assets	4,402,724,932	(2,341,121,903)	2,061,603,029	6,562,096,027	(3,914,368,980)	2,647,727,047
Investments in associates and joint ventures	2,561,802,520	(2,387,705,348)	174,097,172	5,008,234,005	(4,504,098,493)	504,135,512
Deferred tax assets	-	46,583,445	46,583,445	-	589,729,514	589,729,514
Total assets	5,980,408,648	(2,341,121,903)	3,639,286,745	7,183,091,162	(3,914,368,980)	3,268,722,182
Liabilities						
Current liabilities	710,529,787	(38,758)	710,491,029	2,287,442,106	-	2,287,442,106
Current tax liabilities	39,623,718	(38,758)	39,584,960	-	-	-
Derivative liabilities	-	-	-	1,933,562,427	-	1,933,562,427
Non-current liabilities	1,114,410,410	(534,542,330)	579,868,080	919,198,164	(505,315,891)	413,882,273
Deferred tax liabilities	534,542,330	(534,542,330)	-	505,315,891	(505,315,891)	-
Total liabilities	1,824,940,197	(534,581,088)	1,290,359,109	3,206,640,270	(505,315,891)	2,701,324,379
Equity						
Share premium	2,487,313,082	9,313,769	2,496,626,851	2,487,313,082	9,313,769	2,496,626,851
Accumulated other comprehensive loss	(6,975,961)	(1,734,601)	(8,710,562)	(4,009,447)	(3,780,869)	(7,790,316)
Retained earnings	1,509,718,830	(1,814,115,605)	(304,396,775)	1,327,734,757	(3,414,585,988)	(2,086,851,231)
Total equity	4,155,468,451	(1,806,540,815)	2,348,927,636	3,976,450,892	(3,409,053,089)	567,397,803
Total liabilities and equity	₩ 5,980,408,648	(2,341,121,903)	3,639,286,745	7,183,091,162	(3,914,368,980)	3,268,722,182

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

35. Related to the Securities & Futures Commission's ruling and administrative litigation, Continued

(ii) In 2016 and 2015

(In thousands of won)

		2016			2015		
		Before rewriting	Adjustment	After rewriting	Before rewriting	Adjustment	After rewriting
Assets	₩						
Current assets		1,461,425,176	-	1,461,425,176	192,853,895	-	192,853,895
Non-current assets		6,071,590,930	(3,947,716,748)	2,123,874,182	5,767,639,821	(3,982,608,742)	1,785,031,079
Investments in associates and joint ventures		4,944,356,290	(4,519,128,821)	425,227,469	4,837,539,390	(4,533,780,087)	303,759,303
Deferred tax assets		-	571,412,073	571,412,073	-	551,171,345	551,171,345
Total assets		7,533,016,106	(3,947,716,748)	3,585,299,358	5,960,493,716	(3,982,608,742)	1,977,884,974
Liabilities							
Current liabilities		2,476,545,193	-	2,476,545,193	1,911,669,204	-	1,911,669,204
Derivative liabilities		1,874,470,943	-	1,874,470,943	1,820,439,050	-	1,820,439,050
Non-current liabilities		974,091,453	(526,670,614)	447,420,839	1,273,990,355	(581,574,424)	692,415,931
Deferred tax liabilities		526,670,614	(526,670,614)	-	581,574,424	(581,574,424)	-
Total liabilities		3,450,636,646	(526,670,614)	2,923,966,032	3,185,659,559	(581,574,424)	2,604,085,135
Equity							
Share premium		2,487,313,082	9,313,769	2,496,626,851	1,030,620,605	9,313,769	1,039,934,374
Accumulated other comprehensive loss		4,947,005	(3,948,258)	998,747	4,839,666	(3,778,548)	1,061,118
Retained earnings		1,424,706,873	(3,426,411,645)	(2,001,704,772)	1,601,530,283	(3,406,569,538)	(1,805,039,255)
Total equity		4,082,379,460	(3,421,046,134)	661,333,326	2,774,834,157	(3,401,034,318)	(626,200,161)
Total liabilities and equity	₩	7,533,016,106	(3,947,716,748)	3,585,299,358	5,960,493,716	(3,982,608,742)	1,977,884,974

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

35. Related to the Securities & Futures Commission's ruling and administrative litigation, Continued

(iii) In 2014 and 2013

(In thousands of won)

		2014			2013		
		Before rewriting	Adjustment	After rewriting	Before rewriting	Adjustment	After rewriting
Assets	₩						
Current assets		216,838,130	(106,686,799)	110,151,331	156,395,179	(112,364,693)	44,030,486
Non-current assets		1,138,859,825	(57,023,021)	1,081,836,804	503,166,162	16,598,535	519,764,697
Investments in associates and joint ventures		53,921,974	311,196,041	365,118,015	-	166,631,898	166,631,898
Total assets		1,355,697,955	(163,709,820)	1,191,988,135	659,561,341	(95,766,158)	563,795,183
Liabilities							
Current liabilities		236,793,886	(86,282,991)	150,510,895	169,229,512	(40,234,504)	128,995,008
Non-current liabilities		488,171,822	(52,359,864)	435,811,958	105,868,478	(26,406,024)	79,462,454
Total liabilities		724,965,708	(138,642,855)	586,322,853	275,097,990	(66,640,528)	208,457,462
Equity							
Share premium		801,800,313	7,117,379	808,917,692	516,719,342	-	516,719,342
Accumulated other comprehensive loss		9,170,941	(203,819)	8,967,122	(2,341,862)	(136,268)	(2,478,130)
Retained earnings		(314,664,171)	611,394	(314,052,777)	(218,754,883)	416,268	(218,338,615)
Non-controlling interest		32,591,919	(32,591,919)	-	29,405,629	(29,405,629)	-
Total equity		630,732,247	(25,066,965)	605,665,282	384,463,351	(29,125,630)	355,337,721
Total liabilities and equity	₩	<u>1,355,697,955</u>	<u>(163,709,820)</u>	<u>1,191,988,135</u>	<u>659,561,341</u>	<u>(95,766,158)</u>	<u>563,795,183</u>

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

35. Related to the Securities & Futures Commission's ruling and administrative litigation, Continued

(iv) In 2012

(In thousands of won)

		2012		
		Before rewriting	Adjustment	After rewriting
Assets	₩			
Current assets		217,858,628	(141,513,019)	76,345,609
Non-current assets		344,486,626	77,518,098	422,004,724
Investments in associates and joint ventures		-	163,615,991	163,615,991
Total assets		562,345,254	(63,994,921)	498,350,333
Liabilities				
Current liabilities		107,516,998	(34,243,680)	73,273,318
Non-current liabilities		6,366,172	(1,059,834)	5,306,338
Total liabilities		113,883,170	(35,303,514)	78,579,656
Equity				
Share premium		452,386,129	-	452,386,129
Accumulated other comprehensive loss		(990,977)	(97,997)	(1,088,974)
Retained earnings		(83,168,967)	280,000	(82,888,967)
Non-controlling interest		28,873,410	(28,873,410)	-
Total equity		448,462,084	(28,691,407)	419,770,677
Total liabilities and equity	₩	<u>562,345,254</u>	<u>(63,994,921)</u>	<u>498,350,333</u>

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

35. Related to the Securities & Futures Commission's ruling and administrative litigation, Continued

(2) Impact on Financial Performance

If the company loses the administrative litigation, it shall rewrite the statements of comprehensive income as indicated by the Securities and Futures Commission and its impact shall be as follows.

(i) In 2018 and 2017

(In thousands of won)

		2018			2017		
		Before rewriting	Adjustment	After rewriting	Before rewriting	Adjustment	After rewriting
Revenue	₩	535,805,801	-	535,805,801	464,629,334	-	464,629,334
Cost of revenue		(390,645,859)	-	(390,645,859)	(332,829,273)	-	(332,829,273)
Gross profit		145,159,942	-	145,159,942	131,800,061	-	131,800,061
Operating profit		55,678,745	-	55,678,745	65,979,402	-	65,979,402
Equity loss on investments in associates and joint ventures		(129,337,333)	(2,930,169)	(132,267,502)	(129,602,338)	15,030,328	(114,572,010)
Earnings from disposal of investment equity in associates		389,213,672	2,119,447,933	2,508,661,605	-	-	-
Losses on valuation of derivatives		-	-	-	(59,091,484)	-	(59,091,484)
Net profit before Income tax		302,975,837	2,116,517,764	2,419,493,601	(115,994,753)	15,030,328	(100,964,425)
Income tax expenses (benefit)		78,866,642	513,880,872	592,747,514	(19,022,637)	3,204,671	(15,817,966)
Net profit for the period		224,109,195	1,602,636,892	1,826,746,087	(96,972,116)	11,825,657	(85,146,459)
Other comprehensive income (loss)		(2,966,514)	(270,453)	(3,236,967)	(8,956,451)	167,389	(8,789,062)
Total comprehensive income for the period	₩	221,142,681	1,602,366,439	1,823,509,120	(105,928,567)	11,993,046	(93,935,521)

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

35. Related to the Securities & Futures Commission's ruling and administrative litigation, Continued

(ii) In 2016 and 2015

(In thousands of won)

		2016			2015		
		Before rewriting	Adjustment	After rewriting	Before rewriting	Adjustment	After rewriting
Revenue	₩	294,622,021	-	294,622,021	91,278,029	(23,901,796)	67,376,233
Cost of revenue		(268,088,358)	-	(268,088,358)	(114,784,617)	48,453,429	(66,331,188)
Gross profit		26,533,663	-	26,533,663	(23,506,588)	24,551,633	1,045,045
Operating profit (loss)		(30,422,639)	-	(30,422,639)	(203,642,444)	160,536,379	(43,106,065)
Equity loss on investments in associates and joint ventures		(129,222,650)	14,651,266	(114,571,384)	(19,395,735)	(151,241,000)	(170,636,735)
Earnings from disposal of investment equity in subsidiary		-	-	-	4,543,610,535	(4,543,610,535)	-
Losses on valuation of derivatives		(54,031,894)	-	(54,031,894)	(1,820,439,050)	-	(1,820,439,050)
Net profit before Income tax		(230,900,983)	14,651,266	(216,249,717)	2,488,062,610	(4,528,216,321)	(2,040,153,711)
Income tax expenses (benefit)		(54,077,574)	34,493,373	(19,584,201)	583,116,693	(1,132,283,925)	(549,167,232)
Net profit for the period		(176,823,409)	(19,842,107)	(196,665,516)	1,904,945,917	(3,395,932,396)	(1,490,986,479)
Other comprehensive income (loss)		107,338	(169,709)	(62,371)	(8,535,494)	629,489	(7,906,005)
Total comprehensive income for the period	₩	<u>(176,716,071)</u>	<u>(20,011,816)</u>	<u>(196,727,887)</u>	<u>1,896,410,423</u>	<u>(3,395,302,907)</u>	<u>(1,498,892,484)</u>

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

35. Related to the Securities & Futures Commission's ruling and administrative litigation, Continued

(iii) In 2014 and 2013

(In thousands of won)

		2014			2013		
		Before rewriting	Adjustment	After rewriting	Before rewriting	Adjustment	After rewriting
Revenue	₩	105,148,784	(76,130,543)	29,018,241	17,120,845	(17,077,340)	43,505
Cost of revenue		(124,868,406)	41,530,645	(83,337,761)	(54,438,932)	54,399,778	(39,154)
Gross profit		(19,719,622)	(34,599,898)	(54,319,520)	(37,318,087)	37,322,438	4,351
Operating profit (loss)		(119,541,186)	(34,599,898)	(154,141,084)	(154,092,746)	89,212,838	(64,879,908)
Equity loss on investments in associates and joint ventures		(32,679,484)	(35,067,057)	(67,746,541)	-	(73,031,096)	(73,031,096)
Net profit before Income tax		(99,688,402)	3,974,241	(95,714,161)	(148,473,756)	13,024,108	(135,449,648)
Income tax expenses (benefit)		-	-	-	-	-	-
Net profit for the period		(99,688,402)	3,974,241	(95,714,161)	(148,473,756)	13,024,108	(135,449,648)
Other comprehensive income (loss)		11,295,735	149,517	11,445,252	(1,424,072)	34,915	(1,389,157)
Total comprehensive income for the period	₩	<u>(88,392,667)</u>	<u>4,123,758</u>	<u>(84,268,909)</u>	<u>(149,897,828)</u>	<u>13,059,023</u>	<u>(136,838,805)</u>

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2018 and 2017

35. Related to the Securities & Futures Commission's ruling and administrative litigation, Continued

(iv) In 2012

(In thousands of won)

		2012		
		Before rewriting	Adjustment	After rewriting
Revenue	₩	-	-	-
Cost of revenue		-	-	-
Gross profit		-	-	-
Operating profit (loss)		(91,357,686)	51,645,812	(39,711,874)
Equity loss on investments in associates and joint ventures		-	(40,016,257)	(40,016,257)
Net profit before Income tax		(80,241,939)	7,341,692	(72,900,247)
Income tax expenses (benefit)		2,066,460	-	2,066,460
Net profit for the period		(82,308,399)	7,341,692	(74,966,707)
Other comprehensive income (loss)		(1,038,989)	(50,393)	(1,089,382)
Total comprehensive income for the period	₩	<u>(83,347,388)</u>	<u>7,291,299</u>	<u>(76,056,089)</u>

36. Events after the reporting period

In March 2019, the Company and AstraZeneca of the United Kingdom signed an agreement to invest USD 32 million (USD 16 million each) to complete the development of the Rituxan biosimilar of Archigen Biotech Limited, a joint venture.